MOG JEE Main Full Test - 2024 Revised

Total Time: 180 minutes | Total Marks: 300 | Total Questions: 75

GENERAL INSTRUCTIONS

The Question Paper consists of three parts (Physics, Chemistry and Mathematics) with a total of **75** questions.

Each subject is divided into 2 sections:

Section 1: Single Correct Type Questions (4 options, only one correct)

Section 2: Numerical Value Type Questions (answer to be entered as numerical value)

Marking Scheme:

Single Correct: +4 for correct, -1 for wrong, 0 if unattempted.

Numerical Value: +4 for correct, -1 for wrong.

By: C I P H Ξ R

Q.1

A large number of positive charges each of magnitude q are placed along the X-axis at the origin and at every $1\ \text{cm}$ distance in both the directions. The electric flux through a spherical surface of radius $2.5\ \text{cm}$ centred at the origin is

(A) $\frac{5q}{\varepsilon_0}$

(B) $\frac{8q}{\varepsilon_0}$

(C) 0

(D) ∞

Q.2

A straight wire carrying current i is turned into a circular loop. If the magnitude of magnetic moment associated with it in MKS units is M, the length of wire will be

(A) $4\pi iM$

(B) $\sqrt{\frac{4\pi M}{i}}$

(C) $\sqrt{\frac{4\pi i}{M}}$

(D) $\frac{M\pi}{4i}$

Q.3

Assertion (A) Si and GaAs are the preferred materials for solar cells.

Reason (R) Both these materials have energy band gaps much below the energy level corresponding to the maximum solar irradiance ln the solar spectrum.

The correct answer is

(A) (A) is correct but (R) is incorrect.

(B) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(C) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(D) Both (A) and (R) are incorrect.

Q.4

If $200\mathrm{MeV}$ of energy is released in the fission of one nucleus of ${}_{92}^{236}\mathrm{U}$, the number of nuclei that must undergo fission to release an energy of $1000\ \mathrm{J}$ is

(A) 3.125×10^{13}

(B) 6.25×10^{13}

(C) 12.5×10^{13}

(D) 3.125×10^{14}

Q.5

The escape velocity of an object from a planet is $16\mathrm{kms}^{-1}$. If the escape velocity of the object from another planet having twice the density and three times the radius of the planet is $v\sqrt{2}\ \mathrm{ms}^{-1}$, then the value of v is

(A) 12

(B) 48

(C) 18

(D) 36

Q.6

A galvanometer of resistance 40Ω gives a deflection of 10 divisions per mA. There are 50 divisions on the scale. Maximum current that can pass through it when a shunt resistance of 2Ω is connected is

(A) $105\ \mathrm{mA}$

(B) $155\ \mathrm{mA}$

(C) $210\ \mathrm{mA}$

(D) $75\ \mathrm{mA}$

Q.7

To estimate g from $g = 4\pi^2 \frac{L}{T^2}$, error in measurement of L is $\pm 2\%$ and error in measurement of T is $\pm 3\%$. The error in estimated g will be

(A) $\pm 8\%$

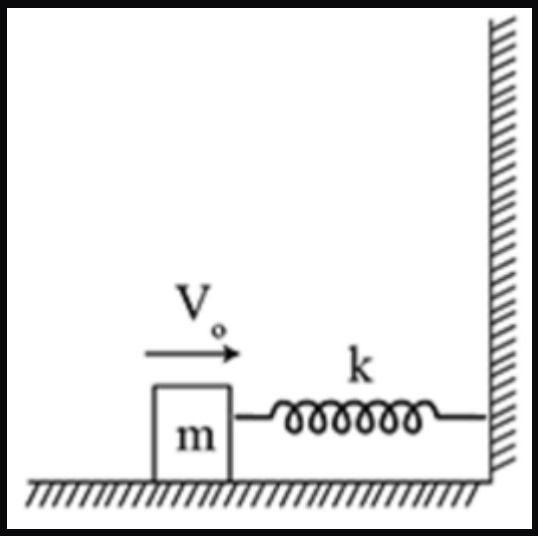
(B) $\pm 5\%$

(C) $\pm 3\%$

(D) $\pm 6\%$

Q.8

In the figure shown, the block of mass m is given velocity V_0 at the spring natural length. The coefficient of friction is given by $\mu = \alpha x$, where α is a positive constant and x is the position of block from the starting point. The block comes to rest for the first time at x equal to :



(A) $V_0\sqrt{\frac{m}{k+\alpha mg}}$

(B) $V_0\sqrt{\frac{m}{\frac{2(k+\alpha mg)}{2m}}}$

(C) $V_0\sqrt{\frac{2m}{k+\alpha mg}}$

(D) $V_0\sqrt{\frac{m}{2k+\alpha mg}}$

Q.9

A $10\mu\mathrm{F}$ capacitor is charged to a potential difference of $50\ \mathrm{V}$ and is connected to another uncharged capacitor in parallel. Now the common potential difference becomes $20\ \mathrm{V}$. What is the capacitance value (in $\mu\mathrm{F}$) of the second capacitor?

(A) $15\mu\mathrm{F}$

(B) $20\mu F$

(C) $25\mu\mathrm{F}$

(D) $5\mu\mathrm{F}$

Q.10

A student is at a distance $16\ \mathrm{m}$ from a bus when the bus begins to move with a constant acceleration of $9\ \mathrm{m\ s^{-2}}$. The minimum velocity with which the student should run towards the bus so as the catch it is $\alpha\sqrt{2}\ \mathrm{ms^{-1}}$. The value of α is

(A) 10

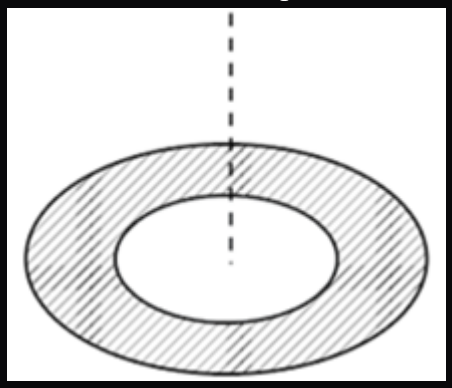
(B) 12

(C) 15

(D) 20

Q.11

A thin disc of radius $b = 2a$ has a concentric hole of radius $'a'$ in it (see figure). It carries uniform surface charge $'\sigma'$ on it. The electric potential on its axis at height $'h'(h << a)$ from its centre is :



(A) $\frac{\sigma a}{2\varepsilon_0}$

(B) $\frac{\sigma a}{4\varepsilon_0}$

(C) $\frac{\sigma a}{3\varepsilon_0}$

(D) $\frac{\sigma a}{8\varepsilon_0}$

Q.12

Two vessels A and B contain oxygen. The volume of B is twice that of A, the pressure of B is thrice that of A and the temperature of B is half of that of A. Then, find the ratio of number of molecules of oxygen in vessels A and B.

(A) 1:3

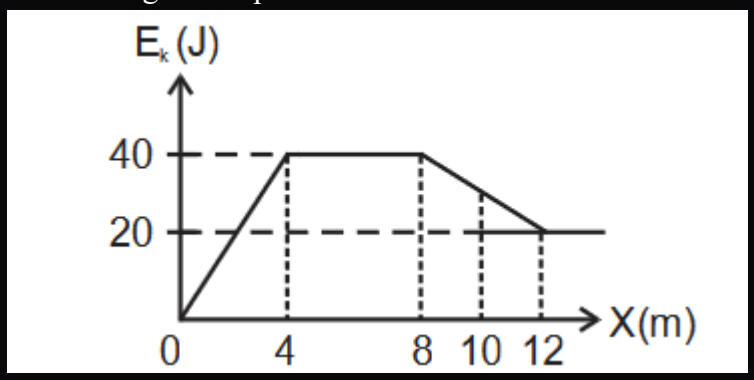
(B) $1:12$

(C) $3:4$

(D) $1:6$

Q.13

The kinetic energy $(\mathrm{E_k})$ of a particle moving along X-axis varies with its position (X) as shown in the figure. The force acting on the particle at $X = 10\ \mathrm{m}$ is



(A) $5\mathrm{i}\hat{N}$

(B) $0\ \mathrm{N}$

(C) $97.5\mathrm{i}\hat{N}$

(D) $-5\mathrm{i}\hat{N}$

Q.14

A string of length $1\ \mathrm{m}$ is fixed at one end with a bob of mass $100\ \mathrm{g}$ and the string makes $\frac{2}{\pi}$ rev s $^{-1}$ around a vertical axis through a fixed point. The angle of inclination of the string with vertical is

(A) $\tan^{-1}(5/8)$

(B) $\tan^{-1}(3/5)$

(C) $\cos^{-1}(8/5)$

(D) $\cos^{-1}(5/8)$

Q.15

The image formed by a convex mirror of focal length $30\ \mathrm{cm}$ is a quarter of the size of the object. The distance of the object from the mirror is

(A) $30\ \mathrm{cm}$

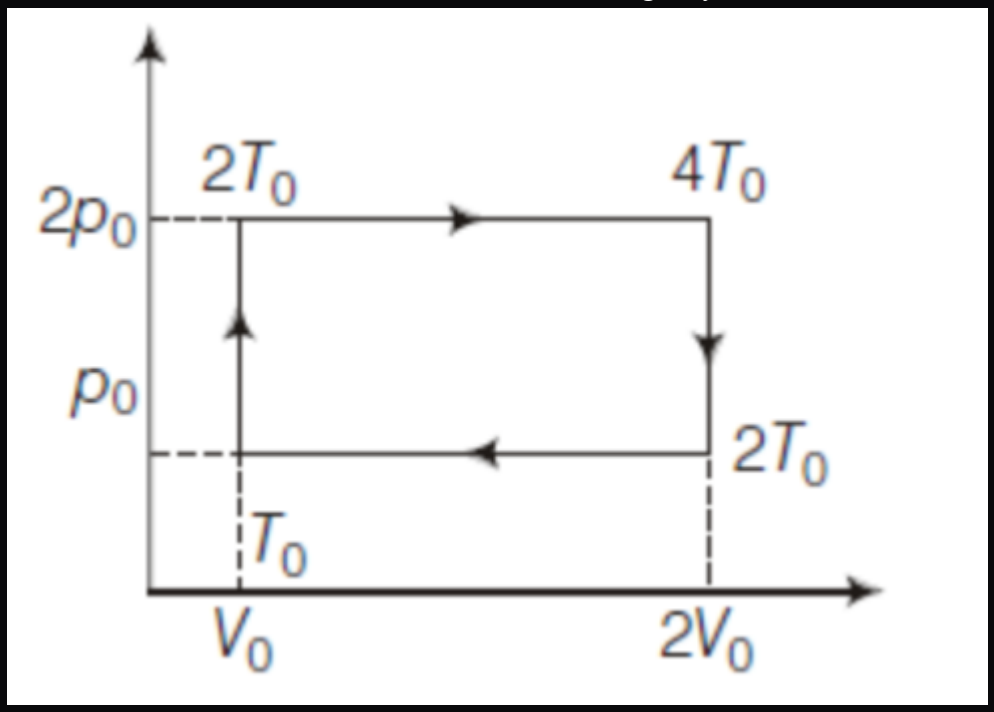
(B) $90\ \mathrm{cm}$

(C) $120\ \mathrm{cm}$

(D) $60\ \mathrm{cm}$

Q.16

The above $p-V$ diagram represents the thermodynamic cycle, operating with an ideal monoatomic gas. The amount of heat, extracted from the source in a single cycle is



(A) p_0V_0

(B) $\left(\frac{13}{2}\right)p_0V_0$

(C) $\left(\frac{11}{2}\right)p_0V_0$

(D) $4p_0V_0$

Q.17

A particle is moving on X-axis has potential energy $U = 2 - 20x + 5x^2$ J along X-axis. The particle is released at $x = -3$. The maximum value of x will be (x is in metre and U is in joules)

(A) 5 m

(B) $3m$

(C) 7 m

(D) 8 m

Q.18

Assertion (A): The upper surface of the wing of an aeroplane is made convex and the lower surface is made concave.
Reason (R): The air currents at the top have smaller velocity and thus less pressure at the bottom than at the top.

(A) Both A and R are true and R is a correct explanation for A

(B) Both A and R are true but R is not a correct explanation for A

(C) A is true, R is false

(D) A is false, R is false

Q.19

Two electric bulbs *A* and *B* are rated as 60 W and 100 W. They are connected in parallel to the same source. Then,

(A) B draws more current than A.

(B) currents drawn are in the ratio of their resistances.

(C) both draw the same current.

(D) A draws more current than B.

Q.20

A convex lens of focal length 25 cm and made of glass with refractive index 1.5 is immersed in water. The absolute change in focal length of the glass is
[Use refractive index of water = $\frac{4}{3}$]

(A) 100 cm

(B) 37.5 cm

(C) 75 cm

(D) 12.5 cm

Q.21

A pair of adjacent coils having a mutual inductance, M. The current in one coil changes from 0 to $16\ \text{A}$ in $0.3\ \text{s}$ and the change of flux linkage with the other coil is $40\ \text{Wb}$. The value of $2M$ is

Q.22

An ionization chamber with parallel conducting plates as anode and cathode has 5×10^7 electrons and the same number of singly-charged positive ions per cm^3. The electrons are moving at $0.4\ \text{m s}^{-1}$. The current density from anode to cathode is 4 μA m^{-2}. The velocity of positive ions moving towards cathode is V, find 100V in m/s.

Q.23

A body is projected horizontally from the top of a tall tower with a velocity of $30\ \text{ms}^{-1}$. At time t_1, its horizontal and vertical components of the velocity are equal and at time t_2, its horizontal and vertical displacements are equal. Then $t_2 - t_1$ is (take, $g = 10\ \text{ms}^{-2}$)

Q.24

YDSE is conducted using light of wavelength 6000 $\mathring{\text{A}}$ to observe an interference pattern. When a film of some material 3.0×10^{-3} cm thick was placed over one of the slits, the fringe pattern shifted by a distance equal to 10 fringe widths. If the refractive index of matrial of the film is K, find 10K.

Q.25

The plates of a parallel plate capacitor are charged upto 200 volts. A dielectirc slab of thickness $4\ \text{mm}$ is inserted between its plates. Then, to maintain the same potential difference between the plates of the capacitor, the distance between the plates is increased by $3.2\ \text{mm}$. The dielectric constant of the dielectric slab is

Q.26

Hydrolysis of proteins gives

(A) β-amino acids

(B) γ-amino acids

(C) δ-amino acids

(D) α-amino acids

Q.27

Find the values of x, y, z in the given balanced equation $\text{H}_2\text{SO}_4 + \text{xHI} \rightarrow \text{H}_2\ \text{S} + \text{yI}_2 + \text{zH}_2\text{O}$

(A) $\text{x} = 3, \text{y} = 5, \text{z} = 2$

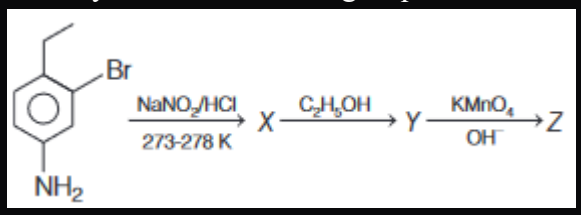
(B) $\text{x} = 4, \text{y} = 8, \text{z} = 5$

(C) $x = 8, y = 4, z = 4$

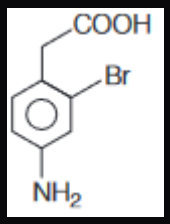
(D) $x = 5, y = 3, z = 4$

Q.28

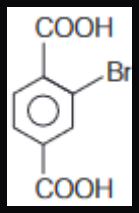
Identify Z in the following sequence of reactions.



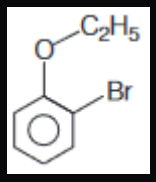
(A)



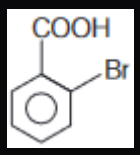
(B)



(C)



(D)



Q.29

Which of the following is not correct about Grignard reagent?

(A) It is a nucleophile

(B) Forms new carbon-carbon bond

(C) Reacts with carbonyl compounds

(D) It is an organomanganese compound

Q.30

The boiling point of water at $1\ \text{atm}$ is 100°C. Which among the following is true for vaporisation of water at 75°C ?

(A) $\Delta G^{\circ}_{\text{vap}} > 0$

(B) $\Delta H^{\circ}_{\text{vap}} < 0$

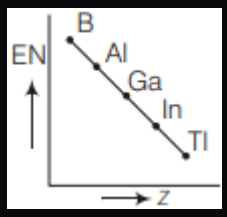
(C) $K_{\text{vap}} = 1$

(D) $\Delta S^{0}_{\text{vap}} < 0$

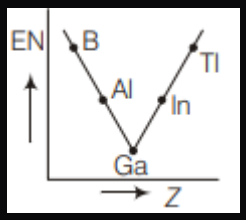
Q.31

Which one of the following correctly represents the variation of electronegativity (EN) with atomic number (Z) of group 13 elements?

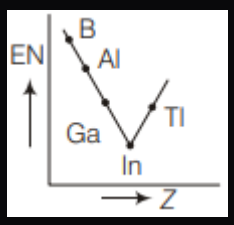
(A)



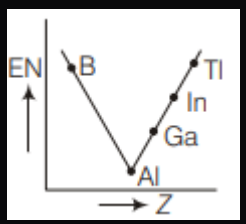
(B)



(C)



(D)



Q.32

A purple colored compound of Manganese (X) decomposes on heating to liberate oxygen and forms compounds of Manganese (Y and Z) compound Z reacts with KOH in presence of potassium nitrate to give compound Y. Compounds X, Y and Z respectively are

(A) $\mathrm{X-KMnO_4\ \ Y-Mn_2O_7\ \ Z-MnO_2}$

(B) $\mathrm{X-K_2MnO_4\ Y-KMnO_4\ Z-Mn_2O_7}$

(C) $\mathrm{X-KMnO_4\ Y-K_2MnO_4\ Z-MnO_2}$

(D) $X - KMnO_4 \quad Y - MnO_2 \quad Z - MnO$

Q.33

$KMnO_4$ reacts with KI, in basic medium to form I_2 and MnO_2. When $250\ mL$ of $0.1M KI$ solution is mixed with $250\ mL$ of $0.02M\ KMnO_4$ in basic medium, what is the number of moles of I_2 formed?

(A) 0.015

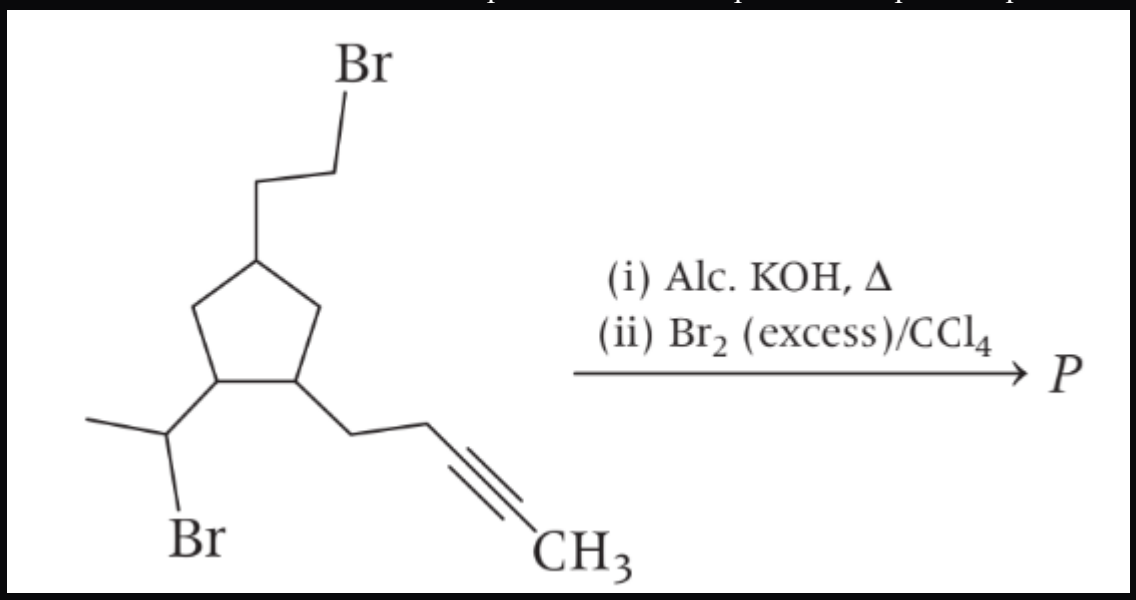
(B) 0.0075

(C) 0.005

(D) 0.01

Q.34

Maximum number of bromine atoms present in the final product P upon complete bromination are



(A) 4

(B) 6

(C) 10

(D) 8

Q.35

Assertion (A) : pKa of phenol is 4.19 and that of benzoic acid is 10
Reason (R) : Phenoxide ion is stabilised by nonequivalent resonance structures whereas benzoate ion by two equivalent resonance structures
The correct option among the following is

(A) A and R are true. R is the correct explanation of A

(B) A and R are true, but R is not the correct explanation for A

(C) A is true R is false

(D) A is false but R is true

Q.36

What are X and Y respectively in the following reaction?

Z-product $\xleftarrow{Y}$ 2-butyne $\xrightarrow{X}$ E-product

(A) Na/NH_3 (liq.) and $Pd/BaSO_4 + H_2$

(B) $Ni/140^\circ C$ and $Pd/BaSO_4 + H_2$

(C) $Ni/140^\circ C$ and Na/NH_3 (liq.)

(D) $Pd/BaSO_4 + H_2$ and Na/NH_3 (liq.)

Q.37

Match the following

	List - I		List - II
a)	Vitamin E	i)	Meat
b)	Vitamin K	ii)	Sunflower oil
c)	Vitamin B_{12}	iii)	Egg white
d)	Vitamin B_2	iv)	Exposure to sunlight
		v)	Green leafy vegetables

The correct answer is

(A)

(a)	(b)	(c)	(d)
iii	ii	iv	i

(B)

(a)	(b)	(c)	(d)
ii	v	i	iii

(C)

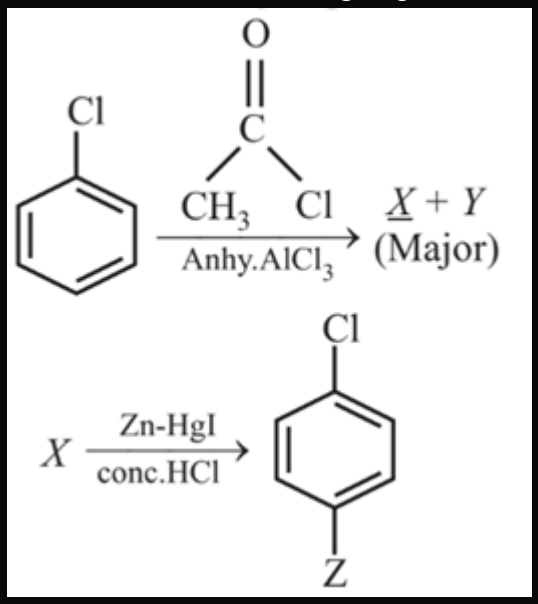
(a)	(b)	(c)	(d)
i	iii	ii	v

(D)

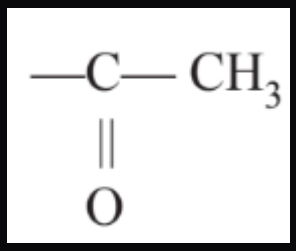
(a)	(b)	(c)	(d)
iv	i	v	ii

Q.38

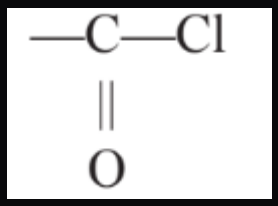
What is $\underline{Z}$ in the following sequence of reactions?



(A)



(B)

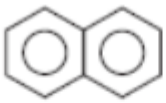
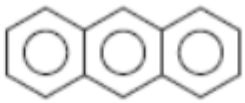
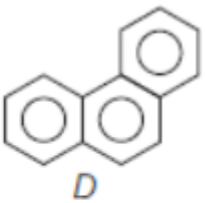


(C) $-CH_3$

(D) $-CH_2-CH_3$

Q.39

The order of stability of aromatic hydrocarbons given below is

Structure	 A	 B	 C	 D
Resonance stabilisation energy in kcal/mol	36	61	84	92

(A) $A < D < B < C$

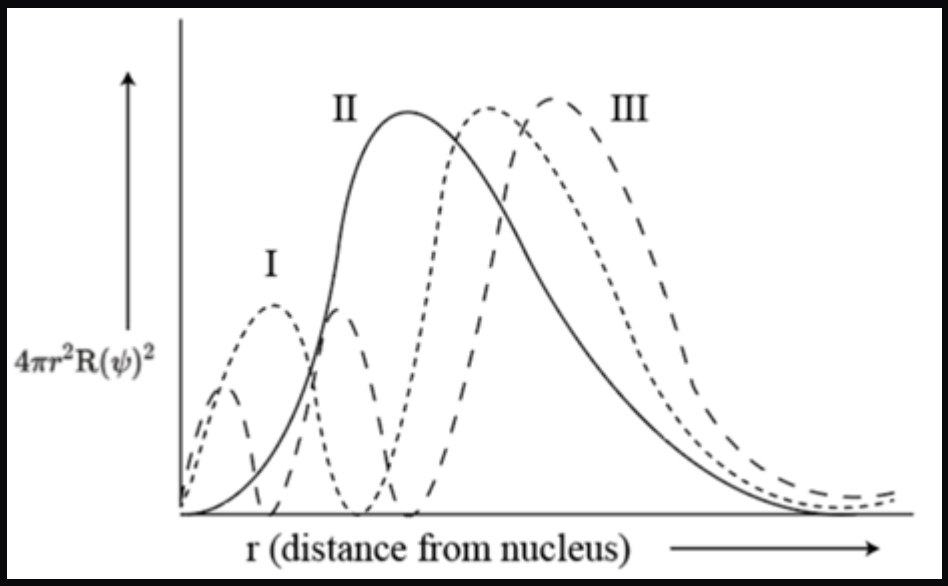
(B) $D < A < B < C$

(C) $B < C < D < A$

(D) $A > D > B > C$

Q.40

Consider the following radial distribution function diagrams. Which of the following has the correct matching of curve and orbital?



(A) $\mathrm{I}(3\ \mathrm{s}), \mathrm{II}(3\mathrm{p}), \mathrm{III}(3\ \mathrm{d})$

(B) $\mathrm{I}(3\ \mathrm{d}), \mathrm{II}(3\mathrm{p}), \mathrm{II}(3\ \mathrm{s})$

(C) $\mathrm{I}(3\ \mathrm{s}), \mathrm{II}(3\ \mathrm{d}), \mathrm{III}(3\mathrm{p})$

(D) I(3p), II(3d), III(3s)

Q.41

The set of amphoteric oxides among $\mathrm{ZnO}, \mathrm{Tl_2O_3}, \mathrm{In_2O_3}, \mathrm{B_2O_3}, \mathrm{PbO}, \mathrm{SnO_2}$ is

(A) $\mathrm{ZnO}, \mathrm{Tl_2O_3}, \mathrm{PbO}$

(B) $\mathrm{ZnO}, \mathrm{PbO}, \mathrm{SnO_2}$

(C) $\mathrm{ZnO}, \mathrm{In_2O_3}, \mathrm{SnO_2}$

(D) $\mathrm{ZnO}, \mathrm{In_2\ O_3}, \mathrm{PbO}$

Q.42

In the Lassaigne's test for the detection of nitrogen in an onganic compound. the appearance of blue coloured compound is due to

(A) ferric ferricyanide

(B) ferrous ferricyanide

(C) ferric ferrocyanide

(D) ferrous ferrocyanide

Q.43

Identify the incorrect order corresponding the mentioned property.

(A) $Al_2O_3 < MgO < Na_2O < K_2O$: Basicity

(B) $NH_3 < PH_3 < AsH_3$: Acidity

(C) $Li^+ < Na^+ < K^+ < Cs^+$: Ionic radius

(D) $Li < Be < B < C$: First ionisation enthalpy

Q.44

Assertion (A) Mg^{2+} and Al^{3+} are isoelectronic but the magnitude of ionic radius of Al^{3+} is less than that in Mg^{2+}.
Reason (R) The effective nuclear charge on the outermost electrons in Al^{3+} is greater than that in Mg^{2+}.
The correct option among the following is

(A) A is true, R is true and R is the correct explanation for A.

(B) A is true, R is true but R is not the correct explanation for A.

(C) A is true but R is false.

(D) A is false but R is true.

Q.45

Sodium acetate was electrolysed by Kolbe's method to form two gases A and B at anode. C and D are formed, when B is heated with regulated supply of O_2 or air in the presence of $(CH_3COO)_2Mn$. C reacts with $NaOH$ to form a salt. A and D are respectively,

(A) CO_2, CH_3COOH

(B) CO_2, H_2O

(C) C_2H_6, H_2O

(D) CO_2, H_2O_2

Q.46

In an experiment to estimate the molecular weight of benzoic acid by elevation in boiling point method, the experimental value of molecular weight was double the actual value. Calculate the degree of association of dimer, if the elevation in Boiling Point is $2^\circ C$.

Q.47

The number of electrons present in bonding and antibonding orbitals in O_2^{2-} is respectively A and B, find $3A - 2B$.

Q.48

At $400\ \mathrm{K}$, in a $1.0\ \mathrm{L}$ vessel, $\mathrm{N_2O_4}$ is allowed to attain equilibrium, $\mathrm{N_2O_4(\ g) \rightleftharpoons 2NO_2(\ g)}$ At equilibrium, the total pressure is $600\ \mathrm{mmHg}$, when 20% of $\mathrm{N_2O_4}$ is dissociated. The value of K_n for the reaction is

Q.49

If the values of Λ_∞ of $\mathrm{NH_4Cl, NaOH}$ and NaCl are 130, 217 and $109 \mathrm{ohm}^{-1}\ \mathrm{cm}^2$ equiv $^{-1}$ respectively, the Λ_∞ of $\mathrm{NH_4OH}$ in ohm $^{-1}\ \mathrm{cm}^2$ equiv $^{-1}$ is

Q.50

Dissolving $120\ \mathrm{g}$ of a compound (mol. wt $= 60$) in $1000\ \mathrm{g}$ of water gave a solution of density $1.12\ \mathrm{g\ mL}^{-1}$. The molarity of solution is

Q.51

Let $A = (0, 4)$ and $B = (2\cos\theta, 2\sin\theta)$, for some $0 < \theta < \frac{\pi}{2}$. Let P divide the line segment AB in the ratio $2 : 3$ internally. The locus of P is

(A) parabola

(B) ellipse

(C) circle

(D) hyperbola

Q.52

Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos t & \sin t \\ 0 & -\sin t & \cos t \end{pmatrix}$

Let $\lambda_1, \lambda_2, \lambda_3$ be the roots of $\det(A - \lambda I_3) = 0$, where I_3 denotes the identity matrix. If $\lambda_1 + \lambda_2 + \lambda_3 = \sqrt{2} + 1$, then the set of possible values of $t, -\pi \le t < \pi$ is

(A) a void set

(B) $\left\{\frac{\pi}{4}\right\}$

(C) $\left\{-\frac{\pi}{4}, \frac{\pi}{4}\right\}$

(D) $\left\{-\frac{\pi}{3}, \frac{\pi}{3}\right\}$

Q.53

The $\lim_{x\to 0} x^8 \left[\frac{1}{x^3}\right]$ (where $[x]$ is greatest integer function) is (Mark incorrect option)

(A) a nonzero real number

(B) a rational number

(C) an integer

(D) zero

Q.54

$\sum_{k=1}^{6} \sin\left(\frac{2\pi k}{7}\right) - i\cos\left(\frac{2\pi k}{7}\right) =$

(A) 1

(B) $-i$

(C) i

(D) -1

Q.55

If [.] represents greatest integer function, then
$\int_{\frac{3\pi}{4}}^{\pi} \left[\sin x + \left[\frac{4x}{\pi}\right]\right] dx =$

(A) $\pi/4$

(B) $\pi/2$

(C) $3\pi/4$

(D) π

Q.56

If $A_i (i = 1, 2, 3, \ldots, n)$ are n independent events with $P\left(A_i\right) = \frac{1}{1+i}$ for each i, then the probability that none of A_i occurs is

(A) $\frac{n-1}{n+1}$

(B) $\frac{n}{n+1}$

(C) $\frac{n}{n+2}$

(D) $\frac{1}{n+1}$

Q.57

The general solution of the differential equation $\left(x^2+2\right) dy + 2xydx = e^x \left(x^2+2\right) dx$ is

(A) $\frac{x}{y} = e^x \left(x^2 + x - 4\right) + c$

(B) $2xy = e^x \left(x^2 - 2x + 4\right) + c$

(C) $\left(x^2+2\right) y = e^x \left(x^2 - 2x + 4\right) + c$

(D) $\left(x^2+2\right)^2 y = e^x \left(x^2 + 2x - 4\right) + c$

Q.58

Let $d(n)$ denotes the number of divisors of n including 1 and itself. Then, $d(225), d(1125)$ and $d(640)$ are

(A) in AP

(B) in HP

(C) in GP

(D) consecutive integers

Q.59

The maximum value of the expression $\left|\sqrt{(\sin^2 x+2a^2)}-\sqrt{(2a^2-1-\cos^2 x)}\right|$, where a and x are real numbers is

(A) $\sqrt{3}$

(B) $\sqrt{2}$

(C) 1

(D) $\sqrt{5}$

Q.60

If $\mathbf{a}=2\hat{\mathbf{i}}+\hat{\mathbf{k}}, \mathbf{b}=\hat{\mathbf{i}}+\hat{\mathbf{j}}+\hat{\mathbf{k}}, \mathbf{c}=4\hat{\mathbf{i}}-3\hat{\mathbf{j}}+7\hat{\mathbf{k}}$, then the vector r satisfying $r \times b=c \times b$ and $r.a=0$ is

(A) $\hat{\mathbf{i}}+8\hat{\mathbf{j}}+2\hat{\mathbf{k}}$

(B) $\hat{\mathbf{i}}-8\hat{\mathbf{j}}+2\hat{\mathbf{k}}$

(C) $\hat{\mathbf{i}}-8\hat{\mathbf{j}}-2\hat{\mathbf{k}}$

(D) $\hat{-\mathbf{i}}-8\hat{\mathbf{j}}+2\hat{\mathbf{k}}$

Q.61

The orthocentre and the centroid of $\triangle ABC$ are $(5,8)$ and $\left(3, \frac{14}{3}\right)$ respectively. The equation of the side BC is $x-y=0$. Given that the image of the orthocentre of a triangle with respect to any side lies on the circumcircle of that triangle, then the diameter of the circumcircle of $\triangle ABC$ is

(A) $\sqrt{10}$

(B) $2\sqrt{10}$

(C) $4\sqrt{10}$

(D) $8\sqrt{10}$

Q.62

Let L be the line parallel to the vector $\sqrt{2}\hat{\mathbf{i}}-5\hat{\mathbf{j}}+3\hat{\mathbf{k}}$ and passing through the point A given by $\hat{\mathbf{i}}+2\hat{\mathbf{j}}-3\hat{\mathbf{k}}$. If the distance between A and a point P on the line L is 18 units, then the position vector of such a point P is

(A) $(1-3\sqrt{2})\hat{\mathbf{i}}+17\hat{\mathbf{j}}-12\hat{\mathbf{k}}$

(B) $(1+3\sqrt{2})\hat{\mathbf{i}}+17\hat{\mathbf{j}}+12\hat{\mathbf{k}}$

(C) $(1+3\sqrt{2})\hat{\mathbf{i}}-17\hat{\mathbf{j}}-12\hat{\mathbf{k}}$

(D) $(1-3\sqrt{2})\hat{\mathbf{i}}-17\hat{\mathbf{j}}+12\hat{\mathbf{k}}$

Q.63

A function $f:\mathbf{R}\to\mathbf{R}$ is such that $f(1)=2$ and $f(x+y)=f(x)\cdot f(y)\forall x,y$. The area (in square units) enclosed by the lines $2|x|+5|y|\leq 4$ expressed interms of $f(1), f(2)$ and $f(4)$ is

(A) $\frac{f(4)}{f(1)+2f(2)}$

(B) $\frac{f(4)}{1+f(2)}$

(C) $\frac{2f(4)}{2f(1)+f(2)}$

(D) $\frac{f(4)}{2f(1)+f(2)}$

Q.64

Consider the sequence of numbers $\alpha_0,\alpha_1,\ldots,\alpha_{\mathrm{n}}$ where $\alpha_0=17.23,\alpha_1=33.23$ and $\alpha_{\mathrm{r}+2}=\frac{\alpha_{\mathrm{r}}+\alpha_{\mathrm{r}+1}}{2}$. Then select incorrect option

(A) $|\alpha_{10}-\alpha_9|=\frac{1}{32}$

(B) $\alpha_0-\alpha_1,\alpha_1-\alpha_2,\alpha_2-\alpha_3,\ldots$ are in G.P.

(C) $\alpha_0-\alpha_2,2\left(\alpha_1-\alpha_2\right),\alpha_1-\alpha_3$ are in H.P.

(D) $|\alpha_{10}-\alpha_9|=|\alpha_8-\alpha_7|$

Q.65

If matrix $A=\begin{bmatrix}1&2\\2&1\end{bmatrix}$ and a non-singular matrix $P=\begin{bmatrix}-1&1\\a&b\end{bmatrix}$ is such that $P^{-1}AP=\operatorname{diag}(-1,3)$, then the sum of all the elements of A^{100} is

(A) 4×3^{100}

(B) 2×3^{100}

(C) 3^{100}

(D) 0

Q.66

Let $X_n=\left\{z=x+iy:|z|^2\leq\frac{1}{n}\right\}$ for all integers $n\geq 1$. Then, $\underset{n=1}{\cap}X_n$ is

(A) a singleton set

(B) not a finite set

(C) an empty set

(D) a finite set with more than one element

Q.67

If the shortest distance between the lines $\mathbf{r} = (3\hat{\mathbf{i}} + 4\hat{\mathbf{j}} - 2\hat{\mathbf{k}}) + t(-\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + \hat{\mathbf{k}})$ and $\mathbf{r} = (\hat{\mathbf{i}} - 7\hat{\mathbf{j}} - 2\hat{\mathbf{k}}) + s(\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 2\hat{\mathbf{k}})$ is equivalent to projection of $\mathbf{P} = -2\hat{\mathbf{i}} + 11\hat{\mathbf{j}}$ on $\mathbf{Q}$ then a possible vector $\mathbf{Q}$ is

(A) $\hat{i} + 5\hat{j} - 3\hat{k}$

(B) $5\hat{i} - \hat{j} + 3\hat{k}$

(C) $-1\hat{\mathrm{i}} + 3\hat{\mathrm{j}} + 5\hat{\mathrm{k}}$

(D) $3\hat{i} + 5\hat{j} - \hat{k}$

Q.68

Let $f : R \to R,\ g : R \to R$ be differentiable functions such that $(fog)(x) = x$. If $f(x) = 2x + \cos x + \sin^2 x$, then the value of $\sum_{n=1}^{99} g(1 + (2n-1)\pi)$ is

(A) 1250π

(B) $(99)^2 \frac{\pi}{2}$

(C) $(99)^2\pi$

(D) 2500π

Q.69

If $f(x) = \log_e\left(e^{2x}\left(\frac{3x+5}{5-3x}\right)^{\frac{2}{3}}\right),\ x \neq \frac{-5}{3}, \frac{5}{3}$, then the value of $\frac{df}{dx}$ at $x = 1$, is

(A) $\frac{5}{4}$

(B) $\frac{7}{4}$

(C) $\frac{11}{4}$

(D) $\frac{13}{4}$

Q.70

If $\int \frac{x^2\left(x\sec^2 x + \tan x\right)}{(x\tan x+1)^2}dx = A\log\left(|x\sin x + \cos x|\right) + B\frac{f(x)}{(x\tan x+1)} + C$, then $f(A+B) =$

(A) 1

(B) 0

(C) -1

(D) 2

Q.71

Let $n = 1! + 4! + 7! + \ldots + 400!$. Then ten's digit of n is

Q.72

Let A be a vertex of the ellipse $S \equiv \frac{x^2}{4} + \frac{y^2}{9} - 1 = 0$ and F be a focus of the ellipse $S' \equiv \frac{x^2}{9} + \frac{y^2}{4} - 1 = 0$. Let P be a point on the major axis of the ellipse $S' = 0$, which divides $\overline{OF}$ in the ratio $2 : 1$ (O is the origin). If the length of the chord of the ellipse $S = 0$ through A and P is $\frac{3\sqrt{101}}{k}$, then $k =$

Q.73

If the general solution of the differential equation $(y - x + 1)dy - (y + x + 2)dx = 0$ is $f(x, y, c) = 0$, then the value of c such that $f(1, 1, c) = 0$ is

Q.74

In a data with 15 number of observations $x_1, x_2, x_3, \ldots, x_{15}, \sum_{i=1}^{15} x_i^2 = 3600$ and $\sum_{i=1}^{15} x_i = 175$. If the value of one observation 20 was found wrong and was replaced by its correct value 40 , then the corrected variance of that data is

Q.75

Let $A = \{1, 2, 3, 4, 5, 6\}$ number of functions f from A to A such that $f(m) + f(n) = 7$, whenever $m + n = 7$ is

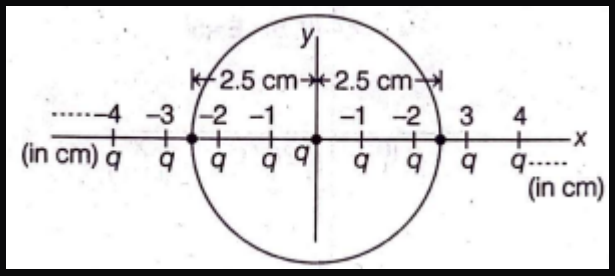
Answers & Solutions

Q.1 Answer:

—

Solution:

The given situation is shown below



From the above figure total charge enclosed by the sphere of radius $2.5 \text{ cm} = 5q$

$\therefore$ According to gauss law, total electric flux passing through spherical surface,

$\phi = \frac{\text{total charge enclosed}}{\varepsilon_0}$

$= \frac{5q}{\varepsilon_0}$

Q.2 Answer:

$\sqrt{\frac{4\pi M}{i}}$

Solution:

Suppose, l be the length of wire.

Hence, $l = 2\pi r \Rightarrow r = \frac{l}{2\pi}$

$\therefore$ Magnetic dipole moment,

$$M = iA$$

$$M = i \cdot \pi r^2 \Rightarrow = i\pi \cdot \left(\frac{l}{2\pi}\right)^2 = i\pi \cdot \frac{l^2}{4\pi^2}$$

$$\Rightarrow M = \frac{il^2}{4\pi}$$

$$l^2 = \frac{4M\pi}{i} \Rightarrow l = \sqrt{\frac{4\pi M}{i}}$$

Q.3 Answer:

(A) is correct but (R) is incorrect.

Solution:

Key Idea The energy band gap of Si is 1.14eV and GaAs is 1.42eV.

Both Si and GaAs are used to make solar cells. In visible spectrum, energy is vary from about 3eV on the violet side to 1.5eV on red side, so visible light can make solar energy.

But radiowaves, microwaves and Infrared waves has less photon energy than the band gap of Si and GaAs.

Hence, the assertion is correct but reason is wrong. Therefore, the correct option is (a).

Q.4 Answer:

3.125×10^{13}

Solution:

Energy released in the fission of one nucleus

$$= 200\text{MeV}$$

$$= 200 \times 1.6 \times 10^{-13}\ \text{J}$$

$$= 3.2 \times 10^{-11}\ \text{J}$$

Energy to be released

$= 1000\ \text{J}$

$3.2 \times 10^{-11}\ \text{J}$ **energy is released due to number of nucleus fissioned** $= 1$

$\therefore 1000\ \text{J}$ **energy is released due to number of nucleus fissioned**

$$= \frac{1 \times 1000}{3.2 \times 10^{-11}}$$

$$= 3.125 \times 10^{13}\ \text{nucleus}$$

Q.5 Answer:

48

Solution:

Escape speed is given by

$$v = \sqrt{\frac{2GM}{R}}$$

$$= \sqrt{\frac{2G \times \frac{4}{3}\pi R^3 \times d}{R}} = \quad (\because M = d \times V).$$

$$\Rightarrow \quad v = \sqrt{\tfrac{8}{3}G\pi R^2 d}$$

Now given for a planet (lets say A)

$$v_1 = 16 = \sqrt{\tfrac{8}{3}G\pi R^2 d}$$

and for planet B,

$$v_2 = \sqrt{\tfrac{8}{3}G\pi (3R)^2 \times 2d}.$$

$$\Rightarrow \quad \frac{v_1}{v_2} = \frac{16}{v_2} = \sqrt{\frac{R^2 \times d}{(3R)^2 \times 2d}}$$

$$\Rightarrow \quad \frac{16}{v_2} = \sqrt{\frac{1}{18}} = \frac{1}{3\sqrt{2}}$$

Hence, $v_2 = 48\sqrt{2}\ \mathrm{m/s} = v\sqrt{2}\ \mathrm{ms}^{-1}$(given)

$$\therefore \quad v = 48$$

Q.6 Answer:

$105\ \mathrm{mA}$

Solution:

Given, galvanometer resistance, $R_G = 40\Omega$

Shunt resistance, $R_\lambda = 2\Omega$

Reading $= 10\mathrm{div/mA}$

and number of divisions, $n = 50$

$\therefore$ Galvanometer current, $I_G = \frac{50}{10} = 5\ \mathrm{mA}$

Let shunt current be I.

$$\text{Since, } \frac{I}{R_G + R_\lambda} = \frac{I_G}{R_\lambda}$$

$$\therefore \quad I = \frac{5 \times (40+2)}{2}$$

$$= 105\ \mathrm{mA}$$

Q.7 Answer:

$\pm 8\%$

Solution:

$\because g = 4\pi^2 \frac{L}{T^2}$

Percentage error in the estimation of g is given as,

$$\frac{\Delta g}{g} \times 100 = \pm\left(\frac{\Delta L}{L} \times 100 + 2\frac{\Delta T}{T} \times 100\right)$$
$$= \pm(2\% + 2 \times 3\%)$$
$$= \pm(2\% + 6\%) = \pm 8\%$$

Q.8 Answer:

$V_0\sqrt{\frac{m}{k+\alpha mg}}$

Solution:

$$\mu mgx + kx = \frac{mvdv}{dx}$$
$$\int_{V_0}^{0} VdV = \int_0^x \left(\mu g + \frac{k}{m}\right) xdx$$
$$x = V_0\sqrt{\frac{m}{K + \alpha mg}}$$

Q.9 Answer:

$15\mu\text{F}$

Solution:

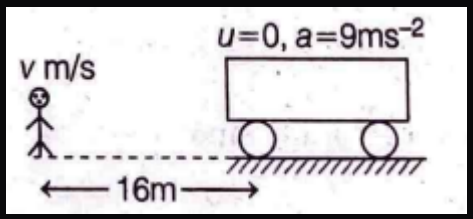
$q_1 = 10 \times 50 = 500\mu\text{C}, C_1 = 10\mu\text{F}, C_2 = ?, q_2 = 0$
As, $V = \frac{q_1+q_2}{C_1+C_2}$
$C_1 + C_2 = \frac{q_1+q_2}{V} = \frac{500+0}{20} = 25\mu\text{F}$
$C_2 = 25 - C_1 = 25 - 10 = 15\mu\text{F}$

Q.10 Answer:

12

Solution:

Let v be the minimum velocity of student so,that he could catch the bus



If student catch the bus in time t, then distance travelled by student in time $t = 16+$ distance travelled by bus in time t.

$\Rightarrow vt = 16 + \left(ut + \frac{1}{2}at^2\right)$

$\Rightarrow vt = 16 + 0 \times t + \frac{1}{2} \times 9 \times t^2$

$\Rightarrow vt = 16 + \frac{9}{2}t^2 \Rightarrow 9t^2 - 2vt + 32 = 0$

The above equation must have real roots. i.e its discriminant ≥ 0

i.e $(2v)^2 - 4 \times 9 \times 32 \geq 0$

$\Rightarrow 4v^2 - 4 \times 288 \geq 0 \Rightarrow v^2 - 288 \geq 0$

$\Rightarrow v^2 \geq 288$

$v \geq 12\sqrt{2}\ \mathrm{m/s}$

Minimum velocity of student to catch the bus $= 12\sqrt{2}\ \mathrm{m/s} = \alpha\sqrt{2}\ \mathrm{m/s}$ **(given)**

$\therefore \alpha = 12$

Q.11 Answer:

$\frac{\sigma a}{2\varepsilon_0}$

Solution:

$r =$ **slant height of elementary ring from points** $P dV_p =$ **electric potential at point** P **due to thin elementary ring of radius** x

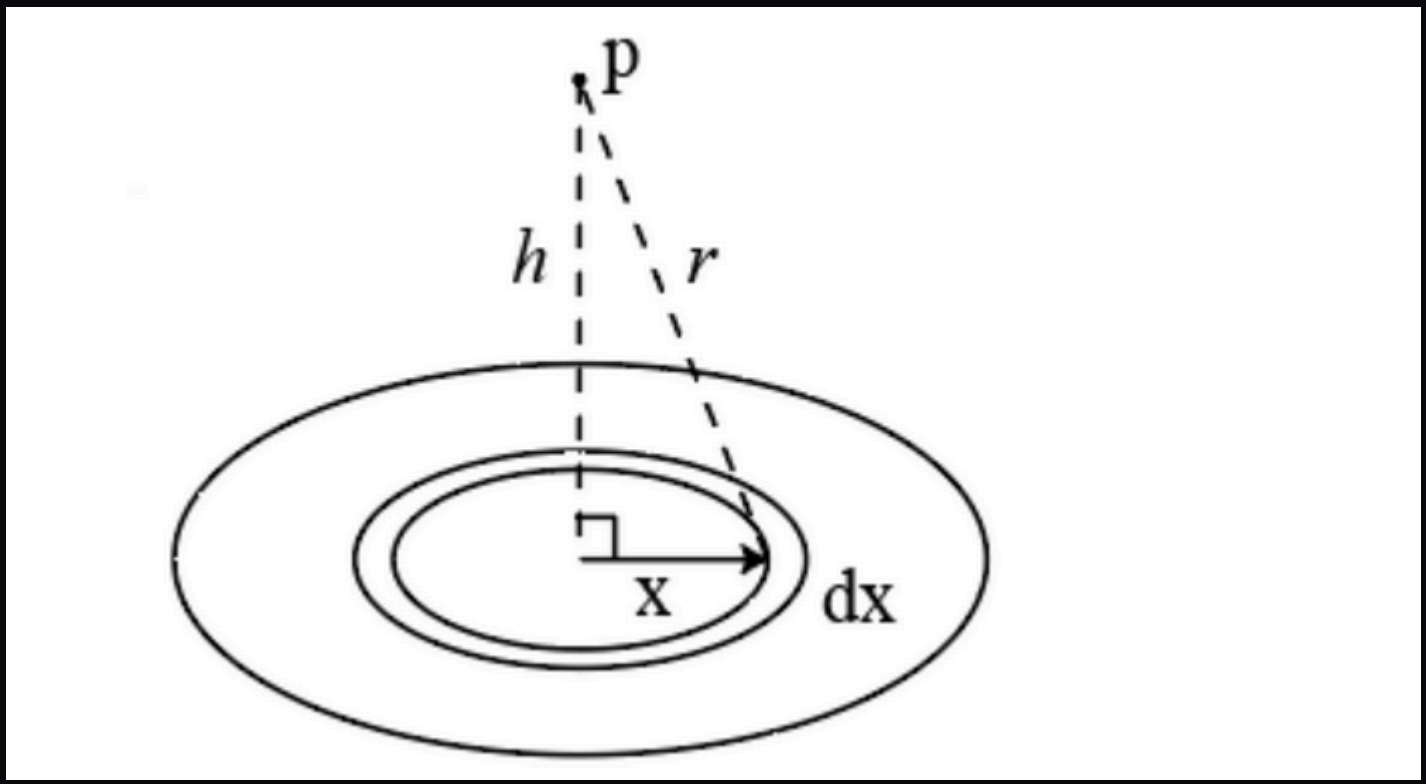
and thickness dx

$dV_p = \frac{1}{4\pi\epsilon_0} \cdot \frac{dQ}{r}, dQ = \sigma 2\pi x \cdot dx$

$V_p = \int dV_p = \frac{1}{4\pi\epsilon_0} \int \frac{\sigma 2\pi x dx}{r} = \frac{\sigma}{2\epsilon_0} \int_a^{2a} \frac{r dr}{r} = \frac{\sigma}{2\epsilon_0} \int_a^{2a} dr = \frac{\sigma a}{2\epsilon_0}$

$r^2 = h^2 + x^2$

$2rdr = 2xdx$



Q.12 Answer:

$1 : 12$

Solution:

For vessels A and B,

$V_B = 2V_A$

$p_B = 3p_A$

$T_B = \dfrac{T_A}{2}$

If N_A and N_B be the number of molecules of oxygens in vessels A and B respectively, then according to ideal gas equation,

$$n_B \times \frac{p_A V_A}{T_A} = n_A \times \frac{p_B V_B}{T_B}$$

$$\Rightarrow \frac{N_B}{M} \cdot \frac{p_A V_A}{T_A} = \frac{N_A}{M} \times \frac{p_B V_B}{T_B}$$

$$\Rightarrow \frac{N_B}{N_A} = \frac{p_B V_B T_A}{p_A V_A T_B}$$

$$\frac{N_B}{N_A} = \frac{3p_A \cdot 2V_A \cdot T_A}{p_A \cdot V_A \cdot \frac{T_A}{2}}$$

$$\frac{N_B}{N_A} = \frac{12}{1} \Rightarrow \frac{N_A}{N_B} = \frac{1}{12} \Rightarrow N_A : N_B = 1 : 12$$

Q.13 Answer:

$-5\hat{\text{i}}N$

Solution:

$\frac{dk}{dx} = \frac{d\left(\frac{1}{2}mv^2\right)}{dx} = \frac{1}{2}m \not{2}v\frac{dv}{dx} = ma = F$

So, Force is slope at $x = 10m$ $\quad F = -\frac{20}{4} = -5\hat{i}$

Q.14 Answer:

$\cos^{-1}(5/8)$

Solution:

$T\cos\theta = mg \ldots\ldots$ (i)

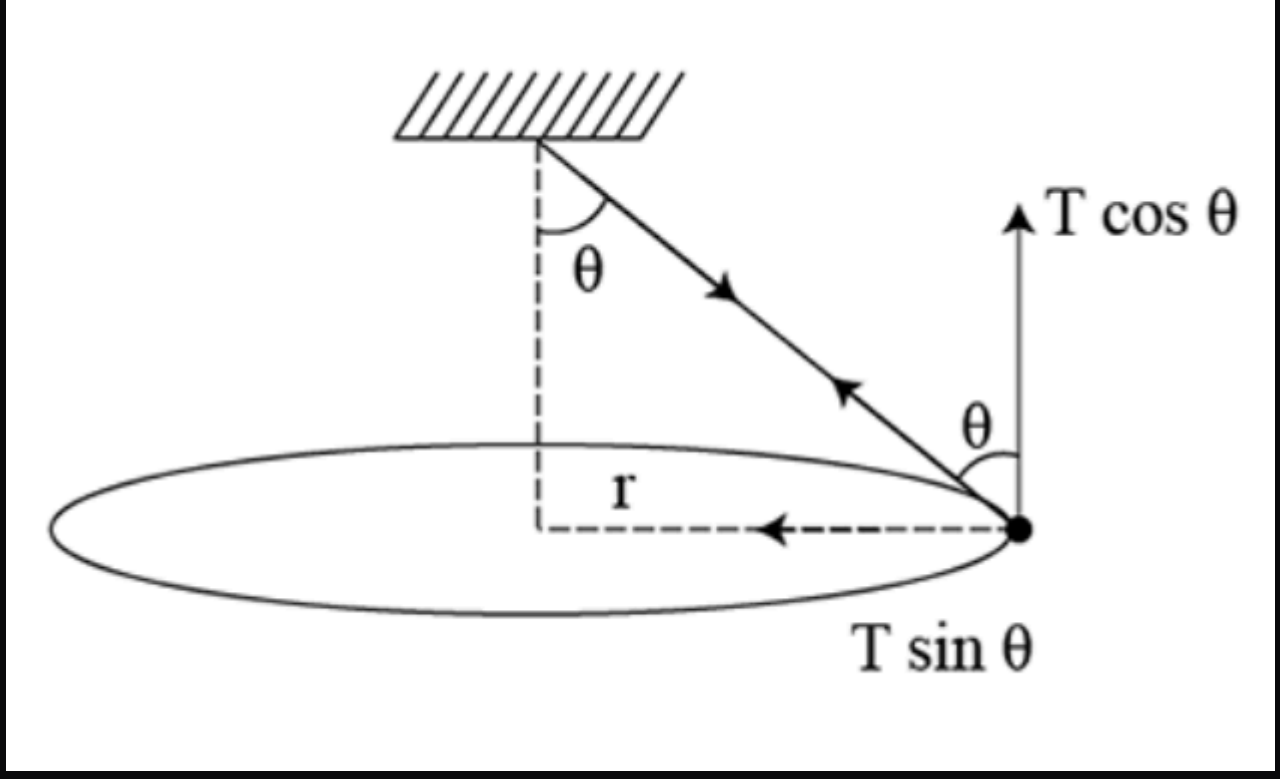
$T\sin\theta = m\omega^2 r \ldots\ldots$ (ii)

$$\text{So } \tan\theta = \frac{\omega^2 r}{g}$$

$$\Rightarrow \frac{\sin\theta}{\cos\theta} = \frac{\omega^2 l \sin\theta}{g} \Rightarrow \cos\theta = \frac{g}{\omega^2 l}$$

$$\Rightarrow \cos\theta = \frac{10}{16 \times 1} = \frac{5}{8}$$

$$\Rightarrow \theta = \cos^{-1}\left(\frac{5}{8}\right)$$



Q.15 Answer:

90 cm

Solution:

Focal length, $f = 30$ cm

$$\mathrm{m} = \frac{-\mathrm{v}}{\mathrm{u}} = \frac{1}{4}$$

$$\mathrm{u} = -4\mathrm{v}$$

$$\frac{1}{\mathrm{f}} = \frac{1}{\mathrm{v}} + \frac{1}{\mathrm{u}} = \frac{1}{\mathrm{v}} - \frac{1}{4\mathrm{v}} = \frac{4-1}{4\mathrm{v}}$$

$$\frac{1}{\mathrm{f}} = \frac{3}{4\mathrm{v}}$$

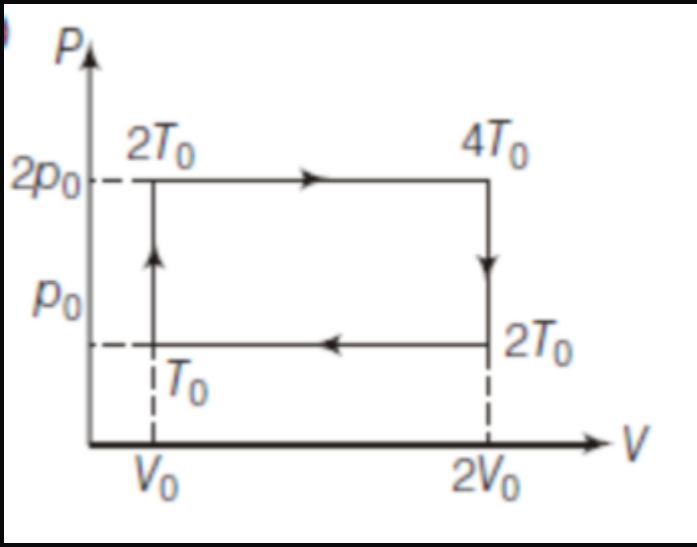
$$\mathrm{f} = \frac{4\mathrm{v}}{3}; \frac{30 \times 3}{4} = \mathrm{v}$$

$$\mathrm{u} = \frac{-4 \times 30 \times 3}{4} = -90 \text{ cm}$$

Q.16 Answer:

$\left(\frac{13}{2}\right) p_0 V_0$

Solution:



Heat supplied,

$H = nC_v\Delta t + nC_p\Delta T$

$= nC_V\,(2T_0 - T_0) + nC_p\,(4T_0 - 2T_0)$

For monoatomic gas, $C_V = \frac{3R}{2}$

and $C_p = \frac{5R}{2}$

$\therefore H = n\frac{3RT_0}{2} + n\left(\frac{5R}{2}\right)2T_0$

$= \frac{3R}{2}nT_0 + 5nRT_0$

$= \frac{13}{2}nRT_0 = \frac{13}{2}p_0V_0$

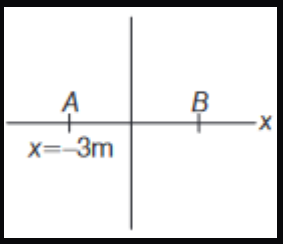
Q.17 Answer:

$7\ \mathrm{m}$

Solution:

Potential energy,

$U = 2 - 20x + 5x^2\,J$



When particle gains whole of potential energy in the form of kinetic energy at $x = -3\ \mathrm{m}$, then it will travel maximum distance till whole of its kinetic energy becomes zero.

i.e., $K = U = 0$

$\Rightarrow \quad 2 - 20x + 5x^2 = 0$

$5x^2 - 20x + 2 = 0$

$\therefore \quad x = \frac{-(-20)\pm\sqrt{(-20)^2 - 4\times5\times2}}{2\times5}$

$= \frac{20\pm\sqrt{360}}{10} = \frac{20\pm18.97}{10}$

$x = 2 \pm 1.9$

For maximum value of x, taking + ve sign

$$x = 2 + 1.9 = 3.9$$

$$\therefore \text{ Total distance } = |-3| + 3.9$$

$$= 6.9 \text{ m} \simeq 7 \text{ m}$$

Q.18 Answer:

A is true, R is false

Solution:

Assertion (A): The upper surface of the wing of an aeroplane is made convex and the lower surface is made concave.

Reason (R): The air currents at the top have smaller velocity and thus less pressure at the bottom than at the top.

Q.19 Answer:

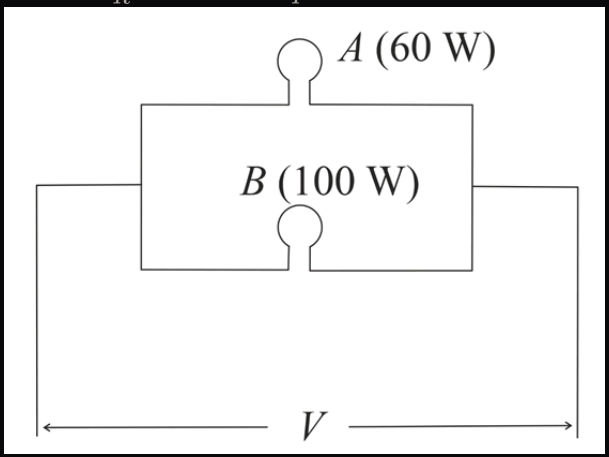
B draws more current than A.

Solution:

Power of the bulbs, when specified without any mentioned testing voltage, voltage can be assumed to be the same.

As $P = \frac{V^2}{R}$ **and** $V =$ **constant**

$\Rightarrow P \alpha \frac{1}{R}$ **or** $R \alpha \frac{1}{P}$



B (100 W) **has less resistance compared to that of** A (60 W)

When resistances are connected in parallel (V **is same)**

As $V = IR$ **,** $V =$**constant (parallel)**

$IR =$ **constant** $\Rightarrow I \alpha \frac{1}{R}$

Smaller the resistance, more is the current drawn. Hence, B **(smaller resistance) draws more current than** A **(larger resistance).**

Q.20 Answer:

75 cm

Solution:

We have

$$\frac{1}{\mathrm{f_{water}}} = ({}^{\mathrm{w}}\mu_{\mathrm{g}} - 1)\left(\frac{2}{\mathrm{R}}\right)$$

$$\text{and, } \frac{1}{\mathrm{f_a}} = ({}^{\mathrm{a}}\mu_{\mathrm{g}} - 1)\left(\frac{2}{\mathrm{R}}\right)$$

$$\text{So, } \frac{\mathrm{f_a}}{\mathrm{f_w}} = \frac{\frac{\mu_g}{\mu_w} - 1}{\mu_g - 1} = \frac{\frac{1}{8}}{\frac{1}{2}} = \frac{1}{4}$$

$$\Rightarrow \mathrm{f_w} = 4\mathrm{f_a} = 100 \text{ cm}$$

$$\text{Now, } \Delta\mathrm{f} = (100 - 25)\mathrm{cm} = 75 \text{ cm}$$

Q.21 Answer:

—

Solution:

Coefficient of mutual induction between 2 coils is given by

$$M = \frac{\text{Flux change in Ind coil}}{\text{Current change in Ist coil}}$$

Given, flux change $= 40$ Wb and

current change $= 16 - 0 = 16$ A

$\Rightarrow$ so, $\quad M = \frac{40}{16} = 2.5\mathrm{H}$

Q.22 Answer:

—

Solution:

In general, the current is,

$i = neAv_d$

Current density is,

$J = \frac{i}{A} = nev_d$

For electrons, the current density is,

$j_e = neV_{d\mathrm{e}}$

$= \left(5 \times 10^7 \times 10^6\right)\left(1.6 \times 10^{-19}\right)(0.4)$

$= 3.2 \times 10^{-6} \text{ A m}^{-2}$

The total current density for the material is,

$J_{total} = J_e + J_{ion}$

$4 \times 10^{-6} = 3.2 \times 10^{-6} + \mathrm{J_{ion}}$

$J_{ion} = 0.8 \times 10^{-6} \text{ A m}^2$

Drift velocity for ions is,

$\left(V_d\right)_{ion} = \frac{J_{ion}}{ne} = \frac{0.8 \times 10^{-6}}{5 \times 10^{13} \times 1.6 \times 10^{-19}} \qquad = 0.1 \text{ m s}^{-1}$

Q.23 Answer:

—

Solution:

As per first condition,

$30 = 10t_1$

$t_1 = 3 \text{ s}$

As per second condition,

$$30t_2 = \frac{1}{2} \times 10t_2^2 = t_2 = 6 \text{ s}$$

$\therefore \quad t_2 - t_1 = 3 \text{ s}$

Q.24 Answer:

—

Solution:

Fringe width, $\beta = \frac{\lambda \mathrm{D}}{\mathrm{d}} \qquad \ldots (i)$

where, D: distance between screen and slit

d: distance between two slits

when a film of thickness t and refractive index μ **is placed over one of the slit, the fringe pattern is shift by distance S and is given by** $\mathrm{S} = \frac{(\mu - 1)\,\mathrm{tD}}{\mathrm{d}} \qquad \ldots (ii)$

Given: $S = 10\,\beta \qquad \ldots (iii)$

From equations (i), (ii) and (iii), we get

$\frac{(\mu-1)+\mathrm{D}}{\mathrm{d}} = 10\,\frac{\lambda \mathrm{D}}{\mathrm{d}}$

$\Rightarrow \mu - 1 = \frac{10\lambda}{t} = \frac{10 \times 6000 \times 10^{-9}\,\mathrm{cm}}{3 \times 10^{-3}\,\mathrm{cm}}$

$\Rightarrow \mu - 1 = 0.2$

$\Rightarrow \mu = 1.2$

Q.25 Answer:

—

Solution:

$$\frac{\varepsilon_0 A}{d} = \frac{\varepsilon_0 A}{d' - t + \frac{t}{K}}$$

$$\Rightarrow \quad d = d' - t + \frac{t}{K} \Rightarrow d' - d = t\left(1 - \frac{1}{K}\right)$$

Here, $d' - d = 3.2 \text{ mm}, t = 4 \text{ mm}$

$$\therefore \quad 3.2 = 4\left(1 - \frac{1}{K}\right) 1 - \frac{1}{K}$$

$$\Rightarrow \quad \frac{3.2}{4} = 1 - \frac{1}{K} \Rightarrow 1 - \frac{1}{K} = \frac{4}{5}$$

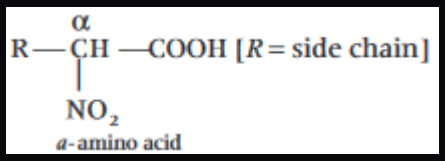
$$\therefore \quad K = 5$$

Q.26 Answer:

α-amino acids

Solution:

Only α-amino acids are obtained by hydrolysis of proteins.



Q.27 Answer:

$x = 8, y = 4, z = 4$

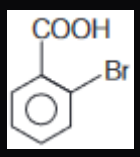
Solution:

Sulphur gains 8 electron in reduction and iodide ion loses 1 electron in oxidation.

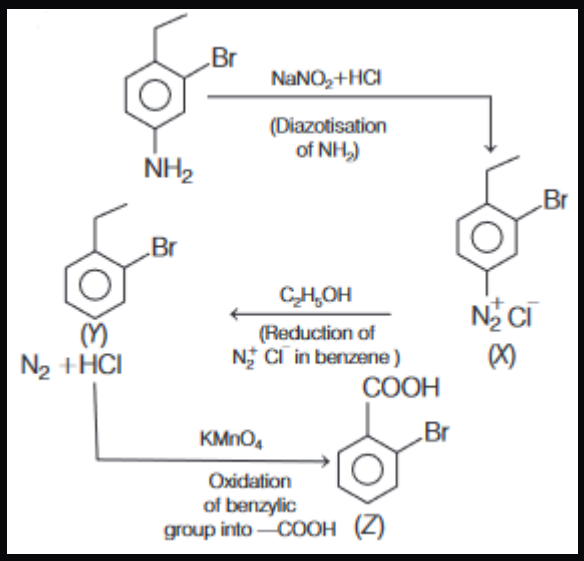
Thus values of $x, \quad y, \quad z$ are $8, 4, 4$ respectively hence the reaction is

$$H_2SO_4 + 8HI \rightarrow H_2\,S + 4I_2 + 4H_2O$$

Q.28 Answer:



Solution:



Thus, option (4) is correct.

Q.29 Answer:

It is an organomanganese compound

Solution:

Grignard reagent $= R - MgX$

Thus, it is an organomagnesium compound.

Q.30 Answer:

$\Delta G^{\circ}_{\text{vap}} > 0$

Solution:

According to standard Gibbs free energy equation

i.e.,

$$\Delta G^{\circ} = \Delta H^{\circ} - T\Delta S^{\circ}$$

$$\Delta G^{\circ} = -T\Delta S^{\circ}$$

$$\Delta G^{\circ} \propto -T$$

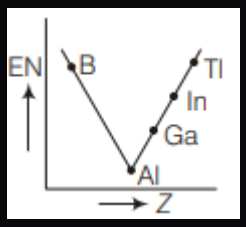
It means ΔG° is inversely proportional to temperature due to negative sign present in Gibbs free energy equation.

Hence, lesser the temperature i.e., 75°C, greater the value of $\Delta^{\circ} G_{\text{vap}}$ $\left(\Delta G^{0}_{\text{vap}} > 0\right)$.

At 100°C, $\Delta G^{0}_{\text{vap}}$ value is lesser than that at temperature 75°C.

$\therefore$ Vaporisation of water at 75°C has $\Delta G^{\circ}_{\text{vap}} > 0$.

Q.31 Answer:

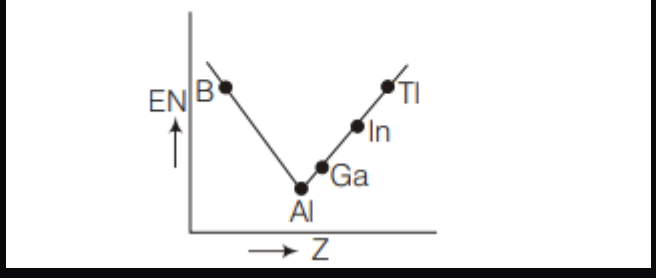


Solution:

For group 13 elements, electronegativity decreases from B to Al. The decrease in electronegativity from B to Al is due to increased size of Al. The remaining three elements are more electro negative than Al and their electroonegativity increases from Ga to TI.

Thus, the correct order of electronegativity of group 13 elements are as follows.

Element	B	Al	Ga	In	Tl
EN (at Pauling scale)	2.0	1.5	1.6	1.7	1.8



Q.32 Answer:

$\text{X} - \text{KMnO}_4\ \text{Y} - \text{K}_2\text{MnO}_4\ \text{Z} - \text{MnO}_2$

Solution:

A purple colored compound of Manganese (X) decomposes on heating to liberate oxygen and forms compounds of Manganese (Y and Z) compound Z reacts with KOH in presence of potassium nitrate to give compound Y. Compounds X, Y and Z respectively are

Q.33 Answer:

0.0075

Solution:

$$\begin{array}{r} \lceil\text{-factor } = 1 \longrightarrow \\ \mathrm{MnO_4^-} + \mathrm{I^-} \rightleftharpoons \mathrm{MnO_2} + \mathrm{I_2} \\ \rfloor^{\rightleftharpoons}_{n\text{-factor} = 3} \end{array}$$

Number of milli equivalents of

$\mathrm{MnO_4^-} = 0.02 \times 3 \times 250 = 15$

Number of milli equivalents of

$\mathrm{I_2} = 0.1 \times 1 \times 250 = 25$

Thus, here $\mathrm{MnO_4^-}$ is limiting reagent.

$\therefore$ Number of milli equivalents of $\mathrm{I_2}$ formed $=$

Number of milli equivalent of $\mathrm{MnO_4^-} = 15$

or number of equivalent of $\mathrm{I_2}$ formed

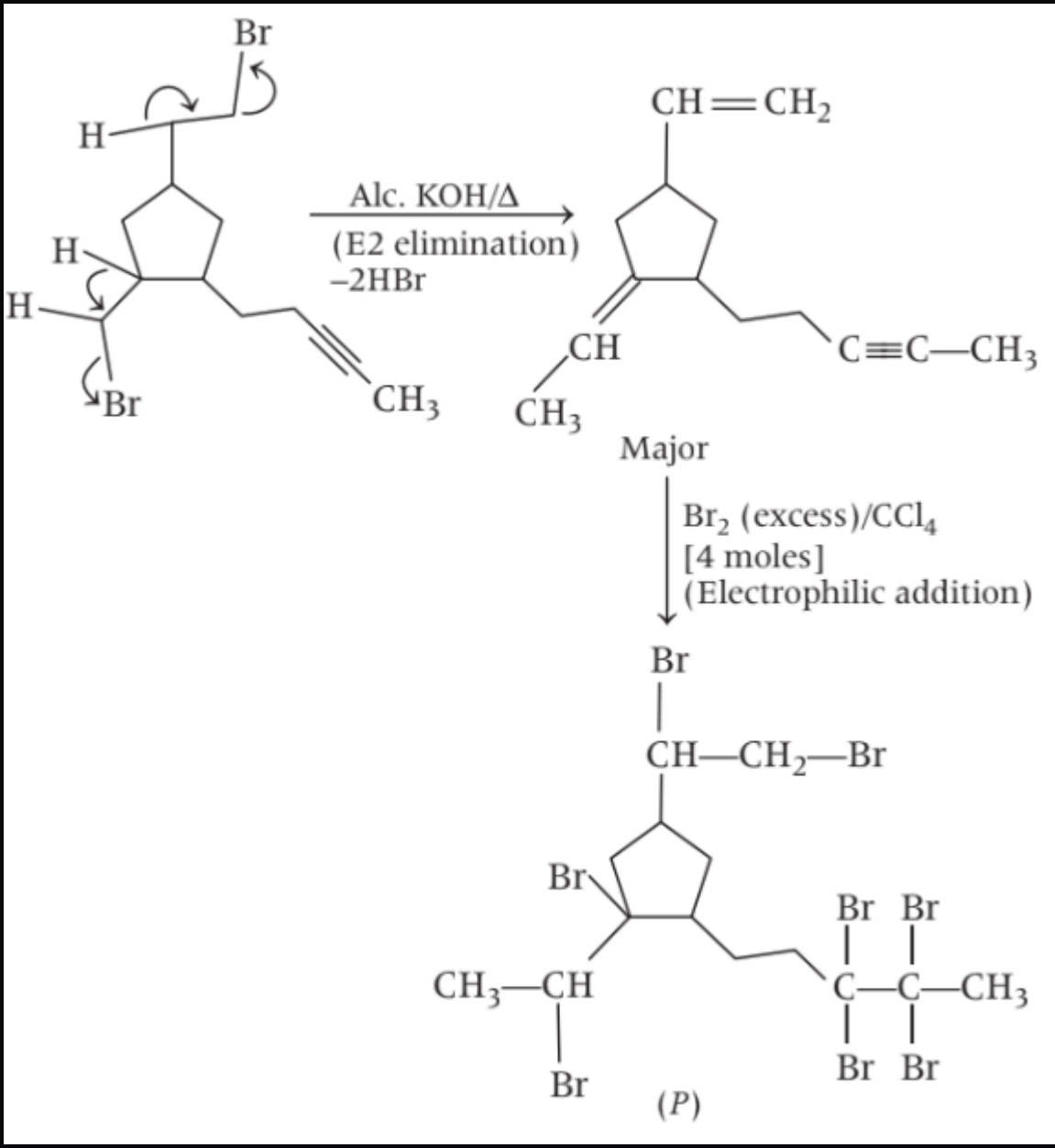
$= \frac{15}{1000} = 0.015$

$$\begin{aligned} \therefore \text{ Number of moles of } \mathrm{I_2} \text{ formed } &= \frac{0.015}{2} \\ &= 0.0075 \end{aligned}$$

Q.34 Answer:

8

Solution:



Q.35 Answer:

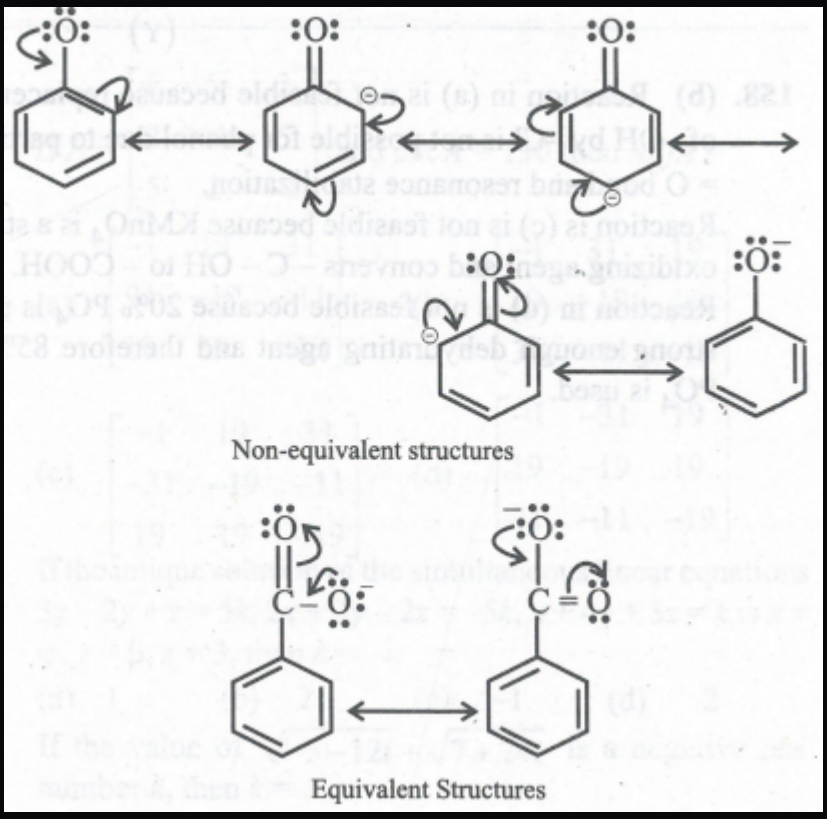
A is false but R is true

Solution:

$pK_a = -\log K_a$ **so as** K_a **increases,** pK_a **decreases. Benzoic acid is a stronger acid than phenol so its** pK_a **will be smaller than that of phenol.**

Thus, Assertion (A) is false.

The resonance structures of phenoxide ion and benzoate ion are given by:-

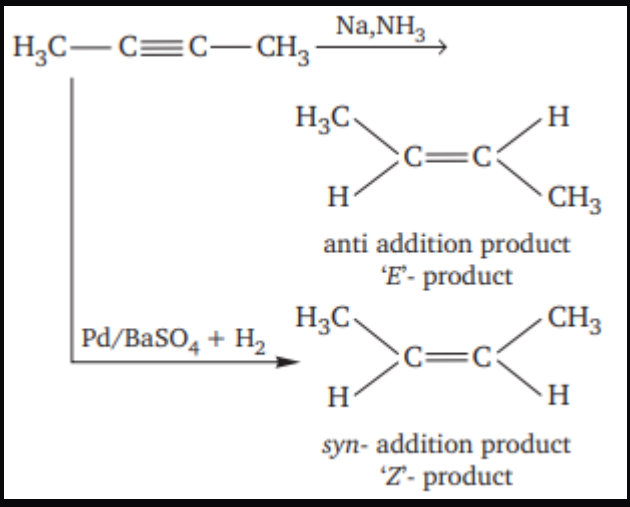


Thus, Reason (R) is true.

Q.36 Answer:

Na/NH_3 (liq.) and $Pd/BaSO_4 + H_2$

Solution:



Hence, reagent X and Y are respectively Na, NH_3 and $Pd/BaSO_4 + H_2$.

Q.37 Answer:

(a)	(b)	(c)	(d)
ii	v	i	iii

Solution:

Missing \end{array}

& \text { Sources } & \text { Deficiency diseases } \ \ \hline \ \text { 1. Vitamin A } &

Fish liver oil, carrots, butter and milk

&

Xeropthalmia (hardening of cornea of eye) Night blindness

\ \

2. Vitamin B_1 (Thjamine)

&

Yeast, milk, green vegetables and cereals

&

Beri beri (loss of appetite, retarded growth)

\ \

3. Vitamin B_2 (Riboflavin)

&

Milk, eggwhite, liver, kidney

&

Cheilosis (fissuring at corners of mouth and lips), digestive disorders and burning sensation of the skin.

\ \

4. Vitamin B_6 (Pyridoxine)

&

Yeast, milk, egg yolk, cereals and grams

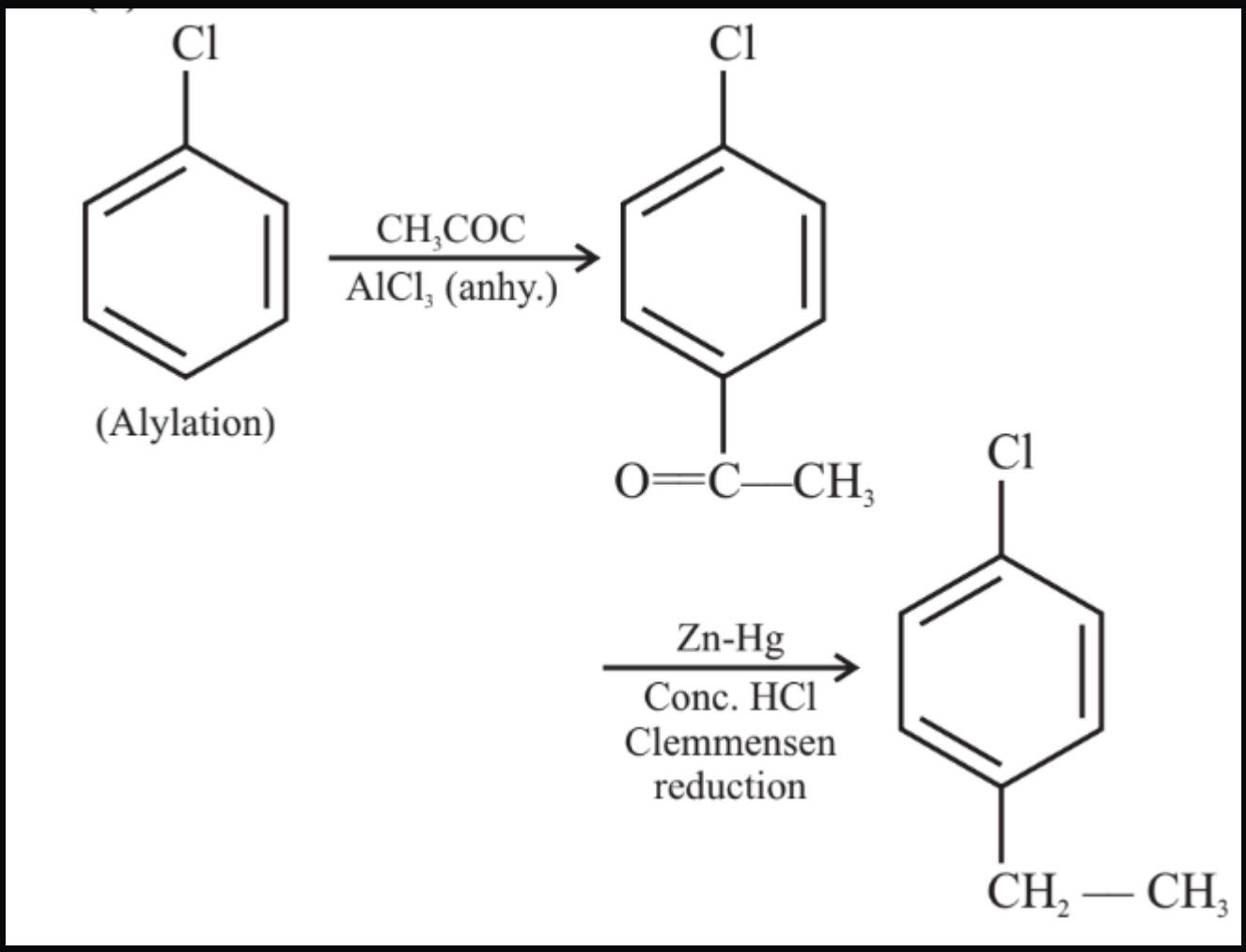
& \text { Convulsions } \ \end{array} \

5. Vitamin B_{12}	Meat, fish, egg and curd	Pernicious anaemia (RBC deficient in haemoglobin)
6. Vitamin C	Citrus fruits, amla and green leafy vegetables	Scurvy (bleeding gums)
7. Vitamin D	Exposure to sunlight, fish and egg yolk	Rickets (bone deformities in children) and osteo-malacia (soft bones and joint pain in adults)
8. Vitamin E	Vegetable oils like wheat germ oil, sunflower oil, etc.	Increased fragility of RBCs and muscular weakness
9. Vitamin K	Green leafy vegetables	Increased blood clotting time

Q.38 Answer:

$-CH_2-CH_3$

Solution:



Q.39 Answer:

$A > D > B > C$

Solution:

Stability of benzene and polycyclic aromatic compounds is determined by resonance energy per ring. Benzene has resonance energy of 36kcal/mol per ring, naphthalene has 30.3kcal/mol, phenanthrene has 30.5kcal/mol and anthracene has 28kcal/mol resonance energy per ring. Stability of aromatic compound is directly proportional to resonance energy per ring. Hence, order of stability : Benzene $(A) >$ phenanthrene $(D) >$ naphthalene $(B) >$ anthracene (C)

Thus, option (4) is correct.

Q.40 Answer:

I(3p), II(3d), III(3s)

Solution:

No of radial node $= \text{n} - \ell - 1.$

$I(3p) - \cdots n = 3$ and $l = 1$. One radial node

$II(3d) - n = 3$ and $\ell = 2$. No radial node

$III(3s) - n = 3$ and $\ell = 0$. Two radial nodes.

All the orbitals provided in choices belong to match. $I(3p), II(3d), III(3s)$

Q.41 Answer:

$\text{ZnO}, \text{PbO}, \text{SnO}_2$

Solution:

Higher oxidation states of metals are acidic. Lower oxidation are basic as they produce bases reacting with water. Amphoteric oxides produce both acids and bases with water or reacts with both acid or bases.

$\text{Tl}_2\text{O}_3, \text{In}_2\text{O}_3 \rightarrow$ basic oxides; $\text{B}_2\text{O}_3 \rightarrow$ acidic oxide $\text{ZnO}, \text{PbO}, \text{SnO}_2 \rightarrow$ acidic oxides

Q.42 Answer:

ferric ferrocyanide

Solution:

To a part of sodium extract, FeSO_4 solution is added and the contents are warmed. A few drops of FeCl_3 solution are then added and resulting solution is acidified with conc. HCl. The appearance of bluish green or Prussian blue colouration confirms the presence of nitrogen. The reactions that occur during this test are

as

$$2NaCN + FeSO_4 \longrightarrow Na_2SO_4 + Fe(CN)_2$$

$$Fe(CN)_2 + 4NaCN \longrightarrow \underset{\text{Sodium ferrocyanide}}{Na_4[Fe(CN)_6]}$$

$$3Na_4[Fe(CN)_6] + 4FeCl_3 \longrightarrow \underset{\substack{\text{Ferric ferrocyanide}\\ \text{(Prussian blue)}}}{Fe_4[Fe(CN)_6]_3} + 12NaCl$$

Q.43 Answer:

$\mathrm{Li} < \mathrm{Be} < \mathrm{B} < \mathrm{C}$: First ionisation enthalpy

Solution:

The first ionisation energy generally increases as we move from left to right across a period. This is due to increasing nuclear charge which results in the strong attraction of outermost electron to the nucleus. So, the correct trend of first ionisation enthalpy is

$\mathrm{Li} < \mathrm{Be} < \mathrm{B} < \mathrm{C}$

Q.44 Answer:

A is true, R is true and R is the correct explanation for A.

Solution:

Higher the electrostatic attraction between nucleus and valence electron, smaller will be the size of the atom/ion.

Electrostatic attraction is given by $Z_{\text{eff}} = Z - S$ where, $Z =$ the number of protons in the nucleus of an atom or ion (the atomic number) and $S =$ shielding from core electrons.

In case of Al^{3+} and Mg^{2+}, they are isoelectronic species and thus have same number of electrons as $1s^2, 2s^2, 2p^6$ isoelectric series of atoms and ions with different numbers of protons (and thus different nuclear attraction, gives) the relative ionic sizes of each atom or ion with respect to atomic number.

Atomic number, $Z_{\mathrm{Al}} = 13$ and $Z_{\mathrm{Mg}} = 12$.

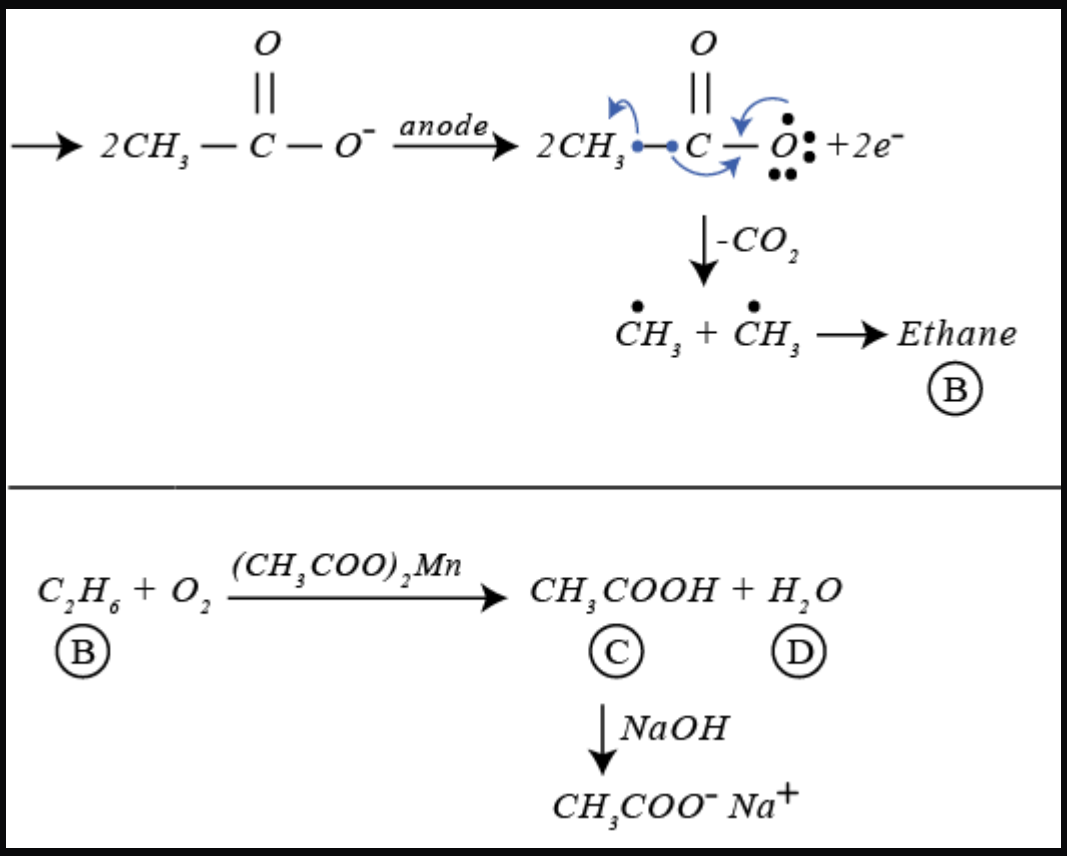
Thus, Al^{3+} has lower ionic radii than Mg^{2+} due to higher nuclear charge.

Hence, A is true, R is true and R is the correct explanation for A.

Q.45 Answer:

$\mathrm{CO_2}, \mathrm{H_2O}$

Solution:



Q.46 Answer:

—

Solution:

$\because$ **According to van't-Hoff volume** (i)

$$(i) = \frac{\text{Molecular mass (calculated)}}{\text{Molecular mass (experimental)}}$$

Given, molecular mass (experimental) is double.

$$\therefore \quad i = \frac{1}{2}$$

Also,

$$\because \quad (i) = (1-\alpha) + \frac{\alpha}{n}$$

where, $\alpha =$ **degree of association and** $n =$ **number of particles associated.**

(Here, $n = 2, \because i = 1/2$ **)**

Therefore,

$$\frac{1}{2} = (1-\alpha) + \frac{\alpha}{2}$$

$$\frac{1}{2} = 1 - \frac{\alpha}{2}$$

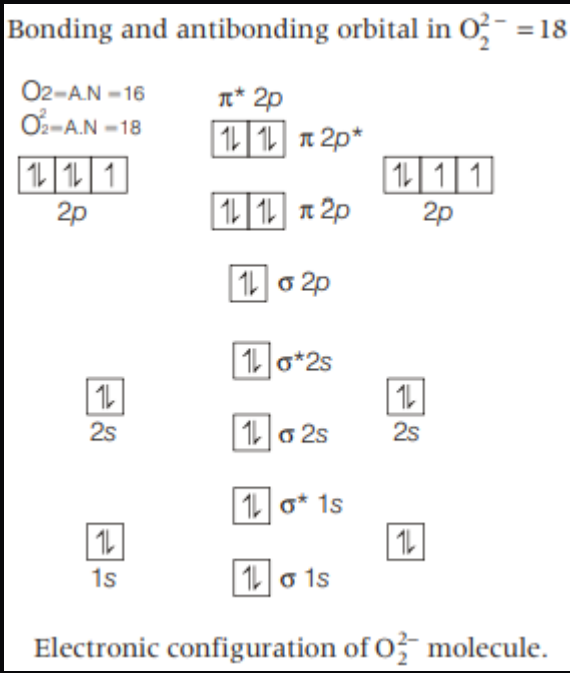
or, $\frac{\alpha}{2} = \frac{1}{2}$

or, $\alpha = 1$

Q.47 Answer:

—

Solution:



Electronic configuration of O_2^{2-} molecule.

$$(\sigma 1s)^2, (\sigma^* 1s)^2, (\sigma 2s)^2, (\sigma^* 2s)^2, \sigma 2p_z)^2,$$
$$(\pi 2p_x^2 = \pi 2p_y^2), (\pi^* 2p_x^2 = \pi^* 2p_y^2)$$

Q.48 Answer:

—

Solution:

$$\begin{array}{lcc} & N_2O_4 \rightleftharpoons & 2NO_2 \\ \text{Initial moles} & 1 & 0 \\ \text{At equilibrium} & (1-0.2) & 2\times 0.2 = 0.8 \end{array}$$

Total moles $= 0.4 + 0.8 = 12$

$$K_P = \frac{(P_{NO_2})^2}{(P_{N_2O_4})} \Rightarrow K_P = \frac{\left(\frac{0.4}{1.2}\times 600\right)^2}{\left(\frac{0.8}{1.2}\times 600\right)} = 100$$

Q.49 Answer:

—

Solution:

$$\Lambda_\infty NH_4Cl = \Lambda_\infty NH_4^+ + \Lambda_\infty Cl^- = 130 \quad \ldots(i)$$
$$\Lambda_\infty NaOH = \Lambda_\infty Na^+ + \Lambda_\infty OH^- = 217 \quad \ldots(ii)$$
$$\Lambda_\infty NaCl = \Lambda_\infty Na^+ + \Lambda_\infty Cl^- = 109 \quad \ldots(iii)$$

On adding Eq. (i) and (ii) and subtracting the Eq. (iii)

$$\Lambda_\infty NH_4^+ + \Lambda_\infty OH^- = 130 + 217 - 109$$

$$\Lambda_\infty NH_4OH = 347 - 109$$

$$= 238 \text{ohm}^{-1} \text{ cm}^2 \text{ equiv}^{-1}$$

Q.50 Answer:

—

Solution:

Given,

density $(d) = 1.12 \text{ g mL}^{-1}$

Mass of solute $(w) = 120 \text{ g}$

$\text{Molar mass of solute } (M) = 60$

$\text{Mass of solvent } (w) = 1000 \text{ g} = 1000 \text{ mL}$

$\therefore \text{ Total mass } = 1000 + 120 = 1120 \text{ g}$

$\because \quad d = \dfrac{\text{Mass}}{\text{Volume } (V)}$

$\therefore \text{ Total volume } (V) = \dfrac{1120}{1.12} = 1000 \text{ mL}$

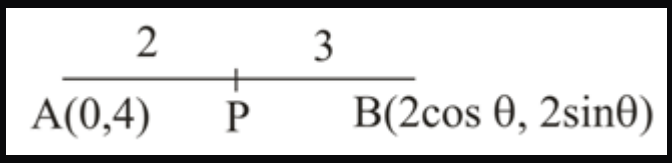
$\because \text{ Molarity } (c) = \dfrac{w}{M} \times \dfrac{1000}{V}$

$\because \text{ Molarity } (c) = \dfrac{w}{m} \times \dfrac{1000}{1000} = \dfrac{120}{60} = 2.0 \text{ m}$

Q.51 Answer:

circle

Solution:



By using section formula:

$\left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}\right)$

$P\left(\frac{4\cos\theta+0}{5}, \frac{4\sin\theta+12}{5}\right) \rightarrow (h, k)$

$\Rightarrow \cos\theta = \frac{5h}{4}, \sin\theta = \frac{5k-12}{4}$

$\Rightarrow \cos^2\theta + \sin^2\theta = 1$

$\Rightarrow \left(\frac{5h}{4}\right)^2 + \left(\frac{5k-12}{4}\right)^2 = 1$

$\Rightarrow (5x)^2 + (5y-12)^2 = 16$

(This is the equation of circle)

Q.52 Answer:

$\left\{-\frac{\pi}{4}, \frac{\pi}{4}\right\}$

Solution:

$$\begin{vmatrix} 1-\lambda & 0 & 0 \\ 0 & \cos t-\lambda & \sin t \\ 0 & -\sin t & \cos t-\lambda \end{vmatrix} = 0$$
$\Rightarrow (1-\lambda)(\cos t-\lambda)^2+\sin^2 t=0$
$\Rightarrow (1-\lambda)\left(\lambda^2-2\lambda\cos t+\cos^2 t\right)+\sin^2 t=0$
$\Rightarrow \lambda^2-2x\cos t+\cos^2 t-\lambda^3+2\lambda^2\cos t+\lambda\cos^2 t+\sin^2 t=0$
$\Rightarrow -\lambda^3+\lambda^2(1+2\cos t)+\lambda\left(\cos^2 t-2\cos t\right)+1=0$
$\lambda_1+\lambda_2+\lambda_3=1+2\,\text{cost}=1+\sqrt{2}$
$\therefore \text{cost}=\frac{1}{\sqrt{2}}\Rightarrow \text{t}=\frac{\pi}{4},-\frac{\pi}{4}$

Q.53 Answer:

a nonzero real number

Solution:

Since $x-1\le[x]\le x$ **for all** $x\in\mathbf{R}$ **so**
$\frac{1}{x^3}-1\le\left[\frac{1}{x^3}\right]\le\frac{1}{x^3}$
$\Rightarrow x^8\left(\frac{1}{x^3}-1\right)\le x^5\left[\frac{1}{x^3}\right]\le x^5$ **for all** x
But $\lim_{x\to0}x^5=0=\lim_{x\to0}\left(x^5-x^8\right)$ **so**
$\lim_{x\to0}x^8\left[\frac{1}{x^3}\right]=0\in\mathbf{I}\subseteq\mathbf{Q}$

Q.54 Answer:

i

Solution:

$\sum_{k=1}^{6}-i\left[\cos\frac{2\pi k}{7}+i\sin\frac{2\pi k}{7}\right]$
$\cos\theta+i\sin\theta=e^{i\theta}$
$(-i)\sum_{k=1}^{6}e^{i\frac{2\pi k}{7}}$
$=(-i)\left[e^{i\frac{2\pi}{7}}+e^{i\frac{4\pi}{7}}+e^{i\frac{6\pi}{7}}+\ldots+e^{i\frac{12\pi}{7}}\right]$
$=-i\left\{e^{i\frac{2\pi}{7}}\frac{\left(1-\left(e^{i\frac{2\pi}{7}}\right)^6\right)}{1-e^{i\frac{2\pi}{7}}}\right\}$ **(sum of GP)**
$=-i\left[\frac{e^{i\frac{2\pi}{7}}-e^{i\frac{14\pi}{7}}}{1-e^{i\frac{2\pi}{7}}}\right]$
$=-i\left[\frac{e^{i\frac{2\pi}{7}}-e^{i2\pi}}{1-e^{i\frac{2\pi}{7}}}\right]\left\{\because e^{i2\pi}=\cos2\pi+i\sin2\pi=1\right.$
$=-i\left[\frac{e^{i\frac{2\pi}{7}}-1}{1-e^{i\frac{2\pi}{7}}}\right]$
$=-i(-1)=i$

Q.55 Answer:

$3\pi/4$

Solution:

Given integral $I = \int_{\frac{3\pi}{4}}^{\pi} \sin x - 1 \left[\frac{4x}{\pi}\right] \cdot dx$

We known that $\sin x$ lies between -1 to 1 and we can split the values of $\sin x$ between -1 to 0 and 0 to 1 .

If $0" \sin x" 1$ then $[\sin x] = 0$ for $x \in [0, \pi]$

Then,

$$\mathrm{I} = \int_{\frac{3\pi}{4}}^{\pi} [\sin x] dx + \int_{\frac{3\pi}{4}}^{\pi} \left[\frac{4x}{\pi}\right] \cdot dx$$

$$\text{If } \frac{3\pi}{4} \leq x \leq \pi \text{ then } \frac{4}{x} \times \frac{3\pi}{4} \leq \frac{4\pi}{\pi} \leq \frac{4}{\pi} \times \pi$$

$$\Rightarrow \quad 3 \leq \frac{4x}{\pi} \leq 4$$

So, $\left[\frac{4x}{\pi}\right] = 3$

$$I = \int_{\frac{3\pi}{4}}^{\pi} (0) dx + 3 \int_{\frac{3\pi}{4}}^{\pi} dx$$

$$I = 0 + 4[x]_{\frac{3\pi}{4}}^{\pi} = 3\left(\pi - \frac{3\pi}{4}\right) = \frac{3\pi}{4}$$

Q.56 Answer:

$\frac{1}{n+1}$

Solution:

The required probability

$= P\left(\bar{A}_1 \cap \bar{A}_2 \cap \ldots \cap \bar{A}_n\right)$

$= P\left(\bar{A}_1\right) P\left(\bar{A}_2\right) \ldots P\left(\bar{A}_n\right)$

$= \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \cdots \cdot \frac{n}{n+1}$

$= \frac{1}{n+1}$

Q.57 Answer:

$\left(x^2+2\right) y = e^x \left(x^2 - 2x + 4\right) + c$

Solution:

$\left(x^2+2\right) dy+2xydx=e^x\left(x^2+2\right) dx$
$\left(x^2+2\right) dy=\left(e^x\left(x^2+2\right)-2xy\right) dx$
$\frac{dy}{dx}=\frac{e^x\left(x^2+2\right)}{x^2+2}-\frac{2x}{x^2+2}y$
$\frac{dy}{dx}+\frac{2x}{x^2+2}y=e^x$
$\mathrm{IF}=e^{\int \frac{2x}{x^2+2}dx}=e^{\ln\left(x^2+2\right)}$
$\mathrm{IF}=x^2+2$
$\therefore y\cdot \mathrm{IF}=\int \mathrm{Q}\cdot(\mathrm{IF})dx+\mathrm{C}$
$y\cdot\left(x^2+2\right)=\int e^x_{\mathrm{II}}\left(x^2+2\right) dx+\mathrm{C}$
$y\left(x^2+2\right)=\left(x^2+2\right)\cdot e^x-\int(2x)\cdot e^x dx+\mathrm{C}$
$y\left(x^2+2\right)=e^x\left(x^2+2-2\right)\int xe^x dx+\mathrm{C}$
$y\left(x^2+2\right)=e^x\left(x^2+2\right)-2\left[x\cdot e^x-\int e^x dx\right]+\mathrm{C}$
$y\left(x^2+2\right)=e^x\left(x^2+2\right)-2xe^x+2e^x+\mathrm{C}$
$y\left(x^2+2\right)=e^x\left(x^2-2x+4\right)+\mathrm{C}$

Q.58 Answer:

in GP

Solution:

Here, $225=3^2\times5^2$
$\Rightarrow d(225)=3\times3=9$
$1126=3^2\times5^3$
$\Rightarrow \quad d(1125)=3\times4=12$
and $640=2^7\times5$
$\Rightarrow \quad d(640)=8\times2=16$
Hence, 9,12 and 16 are in GP.

Q.59 Answer:

$\sqrt{2}$

Solution:

For maximum value
$2a^2-1-\cos^2x=0$
$\therefore \cos^2x=2a^2-1$
$\Rightarrow \sin^2x=1-\cos^2x=\left(2-2a^2\right)$
$\Rightarrow 2a^2+\sin^2x=2$

$\therefore$ **Maximum value of** $\left|\sqrt{(\sin^2 x + 2a^2)} - \sqrt{(2a^2 - 1 - \cos^2 x)}\right|$

$= \left|\sqrt{2} - 0\right| = \sqrt{2}$

Q.60 Answer:

$-\hat{\mathbf{i}} - 8\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$

Solution:

(d) Three vectors a, b **and** c **are given as,** $a = 2\hat{i} + \hat{k}, b = \hat{i} + \hat{j} + \hat{k}$ **and** $c = 4\hat{i} - 3\hat{j} + 7\hat{k}$

Given condition, $\mathrm{r} \times \mathrm{b} = \mathrm{c} \times \mathrm{b}$

$\mathrm{r} \times \mathrm{b} - \mathrm{c} \times \mathrm{b} = 0$

$(\mathrm{r} - \mathrm{c}) \times \mathrm{b} = 0$

It means, $(\mathrm{r} - \mathrm{c}) \| \mathrm{b}$

So, $r - c = \lambda b$

$$\mathrm{r} = \mathrm{c} + \lambda \mathrm{b} \qquad \ldots(\mathrm{i})$$

Also given, $\mathrm{r} \cdot \mathrm{a} = 0$

$(c + \lambda b) \cdot a = 0$ **[using Eq. (i)]**

$(4\hat{i} - 3\hat{j} + 7\hat{k} + \lambda\hat{i} + \lambda\hat{j} + \lambda\hat{k}) \cdot (2\hat{i} + \hat{k}) = 0$

$[(4 + \lambda) \cdot \hat{i} + (-3 + \lambda)\hat{j} + (7 + \lambda)\hat{k}] \cdot (2\hat{i} + \hat{k}) = 0$

$(4 + \lambda) \cdot 2 + (7 + \lambda) \cdot 1 = 0$

$8 + 2\lambda + 7 + \lambda = 0$

$3\lambda = -15$

$\therefore \quad \lambda = -5$

Put the value of λ **in Eq. (i), we get**

$r = 4\hat{i} - 3\hat{j} + 7\hat{k} - 5(\hat{i} + \hat{j} + \hat{k})$

$= 4\hat{i} - 3\hat{j} + 7\hat{k} - 5\hat{i} - 5\hat{j} - 5\hat{k} = -\hat{i} - 8\hat{j} + 2\hat{k}$

Q.61 Answer:

$4\sqrt{10}$

Solution:

The centroid on a non-equilateral triangle divides the line joining orthocentre and circumcentre in $2:1$**, so let the coordinates of circumcentre is** (h, k)**, then**

$$\left(\tfrac{2h+5}{3}, \tfrac{2k+8}{3}\right) = \left(3, \tfrac{14}{3}\right)$$

$\Rightarrow \qquad h = 2k = 3$

And image of orthocentre $(5, 8)$ **with respect to side** $BC, x - y = 0$ **is** $(8, 5)$

So, the radius of circumcircle of $\triangle ABC$ **is**

$\sqrt{(8-2)^2 + (5-3)^2} = \sqrt{40}$

Then diameter $= 4\sqrt{10}$.

Q.62 Answer:

$(1-3\sqrt{2})\hat{\mathbf{i}}+17\hat{\mathbf{j}}-12\hat{\mathbf{k}}$

Solution:

A line L passing through $\hat{\mathbf{i}}+2\hat{\mathbf{j}}-3\hat{\mathbf{k}}$ and parallel to vector $\sqrt{2}\hat{\mathbf{i}}-5\hat{\mathbf{j}}+3\hat{\mathbf{k}}$ is

$\mathbf{r}=\hat{\mathbf{i}}+2\hat{\mathbf{j}}-3\hat{\mathbf{k}}+\lambda(\sqrt{2}\hat{\mathbf{i}}-5\hat{\mathbf{j}}+3\hat{\mathbf{k}})$

Let any point P lie on line is

$(\sqrt{2}\lambda+1,-5\lambda+2,3\lambda-3)$

point $A=\hat{\mathbf{i}}+2\hat{\mathbf{j}}-3\hat{\mathbf{k}}$

Given $PA=18$

$\therefore \quad (\sqrt{2}\lambda+1-1)^2+(-5\lambda+2-2)^2$

$+(3\lambda-3+3)^2=18^2$

$2\lambda^2+25\lambda^2+9\lambda^2=18^2$

$\lambda^2=\dfrac{18^2}{36}=9$

$\lambda=3,-3$

$\therefore \quad$ Point $P=(3\sqrt{2}+1)\hat{\mathbf{i}}-13\hat{\mathbf{j}}+6\hat{\mathbf{k}}$

or $\quad (1-3\sqrt{2})\hat{\mathbf{i}}+17\hat{\mathbf{j}}-12\hat{\mathbf{k}}$

Q.63 Answer:

$\frac{f(4)}{1+f(2)}$

Solution:

Given, $f(x+y)=f(x)\cdot f(y)$

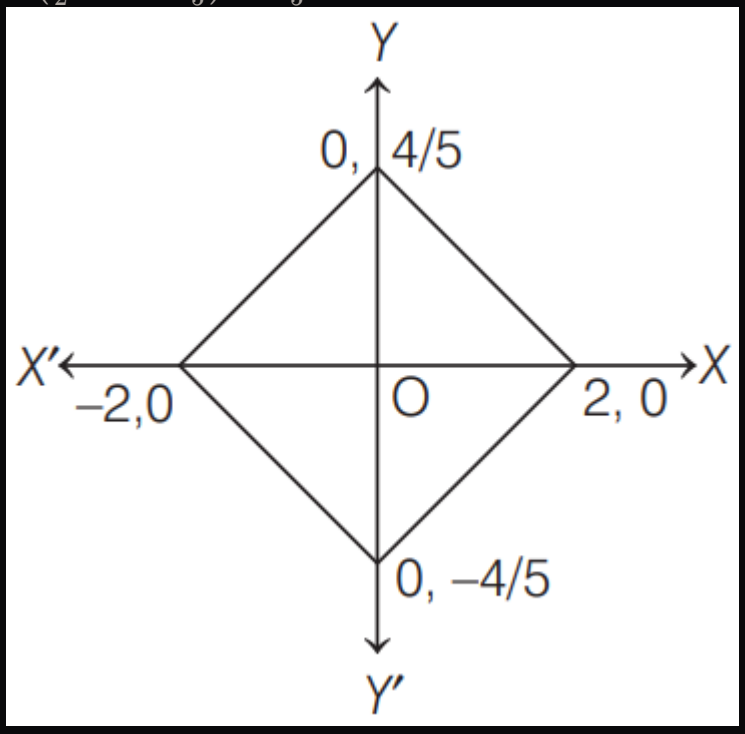
$\therefore \quad f(x)=a^x$

$f(1)=a^1=2\Rightarrow a=2$

$\therefore \quad f(x)=2^x$

Area enclosed by the lines $2|x|+5|y|\leq 4$

$4\left(\frac{1}{2}\times 2\times\frac{4}{5}\right)=\frac{16}{5}$



$=\frac{16}{1+(2)^2}=\frac{(2)^4}{1+(2)^2}=\frac{f(4)}{1+f(2)}$

Q.64 Answer:

$|\alpha_{10}-\alpha_9|=|\alpha_8-\alpha_7|$

Solution:

$2\alpha_{r+2}=\alpha_r+\alpha_{r+1}$

$\Rightarrow 2\left(\alpha_{r+2}-\alpha_{r+1}\right)=\alpha_r-\alpha_{r+1}$

$\Rightarrow \alpha_{r+2}-\alpha_{r+1}=-\frac{1}{2}\left(\alpha_{r+1}-\alpha_r\right)$

$\Rightarrow \alpha_{10}-\alpha_9=-\frac{1}{2}\left(\alpha_9-\alpha_8\right)$

$=\frac{1}{4}\left(\alpha_8-\alpha_7\right)=\ldots$

$=-\frac{1}{2^9}\left(\alpha_1-\alpha_0\right)=\frac{-16}{2^9}=\frac{-1}{32}$

As $\alpha_0-\alpha_1,\alpha_1-\alpha_2,\alpha_2-\alpha_3$, are in G.P.

$\Rightarrow \alpha_0-\alpha_2,2\left(\alpha_1-\alpha_2\right),\alpha_1-\alpha_3$ are in H.P.

(Adding middle term to all terms)

Q.65 Answer:

2×3^{100}

Solution:

$$AP = P\begin{bmatrix} -1 & 0 \\ 0 & 3 \end{bmatrix}$$
$$\Rightarrow \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} -1 & 1 \\ a & b \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ a & b \end{bmatrix}\begin{bmatrix} -1 & 0 \\ 0 & 3 \end{bmatrix}$$
$$\Rightarrow a = 1 = b$$

Now, $A = PDP^{-1}; D = \text{diag}(-1, 3)$

$$\Rightarrow A^{100} = PD^{100}P^{-1} = P\begin{bmatrix} 1 & 0 \\ 0 & 3^{100} \end{bmatrix}P^{-1}$$
$$= \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 3^{100} \end{bmatrix}\begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix}\left(\frac{-1}{2}\right)$$
$$= \frac{1}{2}\begin{bmatrix} 3^{100}+1 & 3^{100}-1 \\ 3^{100}-1 & 3^{100}+1 \end{bmatrix}$$

Q.66 Answer:

a singleton set

Solution:

Given, $X_n = \{z = x + iy : |z|^2 \leq \frac{1}{n}\}$
$= \{x^2 + y^2 \leq \frac{1}{n}\}$
$\therefore$
$X_1 = \{x^2 + y^2 \leq 1\}$
$X_2 = \{x^2 + y^2 \leq \frac{1}{2}\}$
$X_3 = \{x^2 + y^2 \leq \frac{1}{3}\}$
$\dots$
$X_\infty = \{x^2 + y^2 \leq 0\}$

$$\bigcap_{n=1}^{\infty} x_n = X_1 \cap X_2 \cap X_3 \cap \ldots \cap X_\infty$$
$$= \{x^2 + y^2 = 0\}$$

Hence, $\bigcap\limits_{n=1} X_n$ **is a singleton set.**

Q.67 Answer:

$-1\hat{\mathrm{i}} + 3\hat{\mathrm{j}} + 5\hat{\mathrm{k}}$

Solution:

The shortest distance between given lines $\mathbf{r} = (3\hat{\mathbf{i}} + 4\hat{\mathbf{j}} - 2\hat{\mathbf{k}}) + t(-\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + \hat{\mathbf{k}})$ **and** $\mathbf{r} = (\hat{\mathbf{i}} - 7\hat{\mathbf{j}} - 2\hat{\mathbf{k}}) + s(\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 2\hat{\mathbf{k}})$ **is**

$$\begin{vmatrix} 2 & 11 & 0 \\ -1 & 2 & 1 \\ 1 & 3 & 2 \end{vmatrix}$$

$$d = \frac{}{\sqrt{(4-3)^2+(-2-1)^2+(-3-2)^2}}$$

$$= \frac{|2(4-3)-11(-2-1)|}{\sqrt{1+9+25}} = \frac{35}{\sqrt{35}} = \sqrt{35}$$

Now the projection of $\mathbf{p} = -2\hat{\mathbf{i}} + 11\hat{\mathbf{j}}$ on vector $\mathbf{Q} = -\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 5\hat{\mathbf{k}}$ is also $\sqrt{35}$

In option (c) inside of $\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 5\hat{\mathbf{k}}$ put $-\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 5\hat{\mathbf{k}}$ then option (c) is ok

Q.68 Answer:

$(99)^2 \frac{\pi}{2}$

Solution:

Given, $(fog)(x) = x$ **and** $f(x) = 2x + \cos x + \sin^2 x$

So, $g(x)$ **is the inverse of** $f(x)$

$\Rightarrow 1 + (2n-1)\pi = 2x + \cos x + \sin^2 x$

$\Rightarrow 1 + (2n-1)\pi = 2x + \cos x + 1 - \cos^2 x$

$\Rightarrow 2x + \cos x - \cos^2 x = (2n-1)\pi$

Now, $n = 1 \Rightarrow x = \frac{\pi}{2}$

$n = 2 \Rightarrow x = \frac{3\pi}{2}$

$n = 3 \Rightarrow x = \frac{5\pi}{2}$

$\vdots \quad \vdots \quad \vdots$

$n = n \Rightarrow x = (2n-1)\frac{\pi}{2}$

So, $\sum_{n=1}^{99} g(1 + (2n-1)\pi) = \frac{\pi}{2} + \frac{3\pi}{2} + \frac{5\pi}{2} + \ldots\ldots$ **upto 99 terms**

$= \frac{\pi}{2}(1 + 3 + 5 + \ldots\ldots$ **upto 99 terms**$)$

$= (99)^2 \frac{\pi}{2}$

Q.69 Answer:

$\frac{13}{4}$

Solution:

Given,

$f(x) = \log_e\left(e^{2x}\left(\frac{3x+5}{5-3x}\right)^{\frac{2}{3}}\right)$

$\Rightarrow f(x) = \log_e e^{2x} + \log_e\left(\frac{3x+5}{5-3x}\right)^{\frac{2}{3}}$

$\Rightarrow f(x) = 2x + \frac{2}{3}\log_e\left(\frac{3x+5}{5-3x}\right)$

$\Rightarrow f(x) = 2x + \frac{2}{3}(\log_e(3x+5) - \log_e(5-3x))$

Now differentiating the function with respect to x **we get,**

$\frac{df}{dx} = 2 + \frac{2}{3}\left(\frac{1}{3x+5} \times 3 - \frac{1}{5-3x} \times (-3)\right)$

$\Rightarrow \frac{df}{dx} = 2 + 2\left(\frac{1}{3x+5} + \frac{1}{5-3x}\right)$

$\Rightarrow \left(\frac{df}{dx}\right)_{x=1} = 2 + 2\left(\frac{1}{3+5} + \frac{1}{5-3}\right) = 2 + 2 \times \frac{5}{8} = \frac{13}{4}$

Q.70 Answer:

1

Solution:

Let $I = \int x^2 \cdot \frac{x\sec^2 x + \tan x}{(x\tan x+1)^2} dx$

$= x^2\left(-\frac{1}{(x\tan x+1)}\right) - \int 2x\left(-\frac{1}{(x\tan x+1}\right)dx$

$\left[\text{ since } \int \frac{x\sec^2 x+\tan x}{(x\tan x+1)^2} = \int \frac{dt}{t^2} = -\frac{1}{t} = -\frac{1}{(x\tan x+1)}\right]$

$= -\left(\frac{x^2}{x\tan x+1}\right) + \int \frac{2x\cos x}{(x\sin x+\cos x)}dx$

[putting $x\sin x + \cos x = u \Rightarrow (x\cos x + \sin x - \sin x)dx = du$]

$= \frac{-x^2}{(x\tan x+1)} + 2\int \frac{du}{u}$

$= \frac{-x^2}{x\tan x+1} + 2\log|u| + C$

$= \frac{-x^2}{x\tan x+1} + 2\log|x\sin x + \cos x| + C$

$\therefore A = 2$ and $B = -1$ and $f(x) = x^2$

$\therefore f(A+B) = f(2-1) = f(1) = (1)^2 = 1$

Q.71 Answer:

—

Solution:

Given, $n = 1! + 4! + 7! + \ldots + 400!$ $1! = 1, 4! = 24, 7! = 5040, 10! = 3628800$ and further the terms has last two digits $= 00$

So,

$1! + 4! + 7! + \ldots + 400! = \ldots\ldots 65$

Here ten digit is 6 .

Hence, 6 is the answer.

Q.72 Answer:

—

Solution:

We have, $\quad S \equiv \frac{x^2}{4} + \frac{y^2}{9} = 1$

$\therefore \quad A(0, \pm 3) \quad (\text{vertex } = (0, \pm b)$

Again,

$$S' \equiv \frac{x^2}{9} + \frac{y^2}{4} = 1$$

$$\therefore \quad F(\pm\sqrt{5}, 0) \quad (\text{Focus } = (\pm ae, 0)$$

Again P is a point on the major axis of the ellipse $S' = 0$, which divides of in the ratio $2:1$.

$\therefore P = \left(\frac{2x\pm\sqrt{5}+1\times 0}{2+1}, \frac{2\times 0+1\times 0}{2+1}\right) = \left(\frac{\pm 2\sqrt{5}}{3}, 0\right)$

Now, equation of line passing through A and P is given by

$y - 3 = \frac{0-3}{\frac{2\sqrt{5}}{3}-0}(x-0) \Rightarrow y = -\frac{9}{2\sqrt{5}}x + 3$

Now, intersection point of line in Eq. (i) and $S = 0$ are $(0, 3)$ and $\left(\frac{30}{7\sqrt{5}}, \frac{-6}{7}\right)$.

$\therefore$ Length of chord

$$= \sqrt{\left(\frac{30}{7\sqrt{5}} - 0\right)^2 + \left(3 + \frac{6}{7}\right)^2} = \sqrt{\frac{900}{247} + \frac{729}{49}}$$

$$= \sqrt{\frac{4545}{245}} = \sqrt{\frac{909}{49}} = \frac{3\sqrt{101}}{7}$$

$$\therefore \quad k = 7.$$

Q.73 Answer:

—

Solution:

We have,

$$(y - x + 1)dy - (y + x + 2)dx = 0$$

$$ydy - xdy + dy - ydx - xdx + 2dx = 0$$

$$ydy + dy - (xdy + ydx) - xdx + 2dx = 0$$

On integrating, we get

$$\frac{y^2}{2} + y - xy - \frac{x^2}{2} + 2x - C = 0$$

$$f(x, y, c) = \frac{y^2}{2} - \frac{x^2}{2} - xy + 2x + y - C$$

$$f(1, 1, c) = 0$$

$$\therefore \quad \frac{1}{2} - \frac{1}{2} - 1 + 2 + 1 - C = 0 \Rightarrow C = 2$$

Q.74 Answer:

—

Solution:

Given,

$\sum_{i=1}^{15} x_i^2 = 3600, \sum_{i=1}^{15} x_i = 175$

When 20 is replace by 40, then

$$\sum_{i=1}^{15} x_i = 175 - 20 + 40 = 195$$

$$\sum_{i=1}^{15} x_i^2 = 3600 - (20)^2 + (40)^2 = 4800$$

$$\therefore \text{ Corrected variance } = \frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2$$

$$= \frac{4800}{15} - \left(\frac{195}{15}\right)^2 = 320 - 169 = 151$$

Q.75 Answer:

—

Solution:

We have, $A = \{1, 2, 3, 4, 5, 6\}$

$f(m) + f(n) = 7$ of $m + n = 7$

Order of (m, n) is $\{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$

$f(1) + f(6) = 7, f(2) + f(5) = 7, f(3) + f(4) = 7$

$f(1) + f(6) = 7$ has $36 \times 2 = 72$ functions

Similar for $f(2) + f(5) = 7$ and $f(3) + f(4) = 72$

Total number of functions $= 72 + 72 + 72 = 216$